INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

BOOTS, HOT WEATHER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers hot weather, combat boots that have a leather and nylon duck upper and direct molded sole. The type I boot contains a built-in spike resistant insole, and the type II boot has an insole which is resistant to heat transfer.

1.2 Classification. The boots shall be of the following types, sizes, and widths as specified (see 6.2). The schedule of sizes is as listed in table I.

Type I - Black, hot, wet
Type II - Tan, hot, dry

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5019 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8430

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TABLE I. Schedule of sizes

Width	3	3-1/2	4	4-1/2	5	5-1/2	6	6-1/2	7	7-1/2	8	8-1/2
XN	-	-	-	-	X	X	X	X	X	X	X	X
N	-	-	X	X	X	X	X	X	X	X	X	X
R	-	-	X	X	X	X	X	X	X	X	X	X
W	-	-	X	X	X	X	X	X	X	X	X	X
XW	X	X	X	X	X	X	X	X	X	X	X	X

Width	9	9-1/2	10	10-1/2	11	11-1/2	12	12-1/2	13	13-1/2	14
XN	X	X	X	X	X	X	X	X	X	X	X
N	X	X	X	X	X	X	X	X	X	X	X
R	X	X	X	X	X	X	X	X	X	X	X
W	X	X	X	X	X	X	X	X	X	X	X
XW	X	X	X	X	X	X	X	X	X	X	X

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

A-A-203 - Paper, Kraft, Untreated
A-A-1249 - Paper, Wrapping, Tissue
V-L-61 - Laces, Nylon
V-T-285 - Thread, Polyester
V-T-295 - Thread, Nylon
KK-I-570 - Insole, Footwear, Leather, Cattlehide
QQ-B-613 - Brass, Leaded and Nonleaded: Flat Products (Plate, Bar, Sheet, and Strip)
TT-C-490 - Cleaning Methods for Ferrous Surfaces and Pretreatments for Organic Coatings
DDD-L-20 - Label: For Clothing, Equipage, and Tentage, (General Use)
DDD-T-86 - Tape, Textile; Cotton, General Purpose (Unbleached, Bleached, or Dyed)

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GGG-N-350	-	Nippers and Pincers
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-F-320	-	Fiberboard, Corrugated and Solid, Sheet-Stock (Container Grade) and Cut Shapes

MILITARY

MIL-I-1821	-	Insoles, Footwear (Ventilating Type)
MIL-L-3122	-	Leather, Cattlehide, for Footwear Uppers, Chrome Tanned, Fatliquored
MIL-T-5038	-	Tape, Textile and Webbing, Textile, Reinforcing, Nylon
MIL-S-5059	-	Steel, Corrosion-Resistant (18-8), Plate, Sheet and Strip
MIL-L-10867	-	Leather, Cattlehide, Gusset, Chrome Tanned, Fatliquored
MIL-C-13924	-	Coating, Oxide, Black, for Ferrous Metals
MIL-W-17337	-	Webbing, Textile, Woven Nylon
MIL-L-35078	-	Loads, Unit: Preparation Of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-C-41814	-	Counter, Footwear
MIL-L-41815	-	Lasts, Footwear, Combat, U.S., MIL-5
MIL-C-43734	-	Cloth, Duck, Textured Nylon
MIL-C-43956	-	Cloth, Twill, Cotton, 10.0 Ounces (339 g)

STANDARDS

FEDERAL

FED-STD-191	-	Textile Test Methods
FED-STD-311	-	Leather, Methods of Sampling and Testing
FED-STD-595	-	Colors Used in Government Procurement
FED-STD-601	-	Rubber: Sampling and Testing
FED-STD-751	-	Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads
MIL-STD-731	-	Quality of Wood Members for Containers and Pallets

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

2-1-962	-	Panama Tread Sole for Hot Weather; Boots; (Sole Dimensions)
2-1-1635	-	Speed Lace Assembly

(Copies of drawings are available from the U.S. Army Natick Research, Development, and Engineering Center, ATTN: STRNC-UX, Natick, MA 01760-5017).

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 36	-	Brass Plate, Sheet, Strip, and Rolled Bar
D 518	-	Rubber Deterioration - Surface Cracking
D 573	-	Rubber Deterioration in an Air Oven
D 792	-	Plastics by Displacement, Specific Gravity (Relative Density) and Density of
D 1052	-	Rubber Deterioration - Cut Growth Using Ross Flexing Apparatus
D 1055	-	Latex Foam, Flexible Cellular Materials
D 1149	-	Rubber Deterioration - Surface Ozone Cracking in a Chamber (Flat Specimens)
D 1630	-	Rubber Property - Abrasion Resistance (NBS Abrader)
D 2240	-	Rubber Property - Durometer Hardness
D 3574	-	Standard Methods of Testing Flexible Cellular Materials - Slab, Bonded, and Molded Urethane Foams
D 3575	-	Method of Testing Flexible Cellular Materials Made From Olefin Plastics
E 18	-	Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
E 96	-	Water Vapor Transmission of Materials in Sheet Form

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Standard sample. The standard sample for color and sueded flesh appearance shall be a swatch of cattlehide upper (see 6.4).

3.3 Materials. It is encouraged that recycled material be used, except for textile, leather, or rubber components, as long as it meets the requirements of this specification.

3.3.1 Upper leather.

3.3.1.1 Type I. The leather for the vamps, counterpockets, quarter front stays, and eyelet stays shall be smooth plated, full grain, or corrected grain side leather conforming to treatment A of MIL-L-3122, except that the thickness shall be as specified for the cut parts in table II. The break for the vamp shall be not more than a break pattern of 4 when examined as specified in 4.4.3. The color shall be black. All finishes applied by the boot manufacturer and tanner shall be compatible with one another to ensure a long-lasting low-gloss finish. The tear strength for the inside counter pockets, front edge stays and eyelet stays shall not apply.

3.3.1.2 Type II. The leather for the vamps, counterpockets, ankle reinforcements, front edge stays and eyelet stays shall conform to treatment A of MIL-L-3122, except that:

a. The flesh surface shall be sueded so as to produce a fine uniform nap (see 3.2).

b. The grain surface shall be lightly buffed to remove the surface of the grain only (see 3.2).

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c. The color shall be desert tan and shall be obtained by drum dyeing (see 3.2). The dye shall penetrate into the leather from both the flesh and grain of the total thickness. The finish requirements shall not apply.

d. Shank and pocket areas, along with any other location on the cattlehide side which have loose flesh or raggy fibers shall not be used.

e. The grain side of the leather shall be in contact with the test pad when tested for staining.

f. The tear strength requirements for the ankle reinforcement, front edge stays, inside counter pockets, and eyelet stays shall not apply.

g. At least 80 percent of the ankle reinforcement/front edge stay leather test specimens shall have stitch tearing strength values of not less than 30 pounds when tested in accordance with MIL-L-10867.

h. The thickness shall be as specified for the cut parts in table II.

TABLE II. Thickness of upper leather.

Part	Thickness (ounces)			
	If not split		For split parts	
	Minimum	Maximum	Minimum	Maximum
Vamp	5	6	-	-
Outside counterpocket	4	6	4	6
Inside counterpocket	3	4	3-1/2	4
Front edge stays	3	4	3-1/2	4
Eyelet stays	3	4	3-1/2	4
Ankle reinforcement (type II only)	3	4	-	-

3.3.1.3 Leather insoles. The leather insoles shall consist of two thicknesses: a 4-iron insole conforming to type I, tannage a or b, class 1 of KK-I-570 and a 3-iron insole conforming to type I or III, tannage a or b, class 1 of KK-I-570. Slaughter cuts not penetrating more than one-third of the thickness of the insole may be permitted in the 3-iron insole. Leather insoles split from a single unit shall conform to the above specified thicknesses after splitting. The insoles for this boot do not contain a channel area.

3.3.1.4 Counters. The counters shall conform to the requirements of MIL-C-41814 for Boots, Hot Weather. The counter size schedule shall be as follows:

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Size	XN	N	R	W	XW
1	-	3 - 3-1/2	-	-	-
2	5 - 6-1/2	4 - 5-1/2	3 - 4	-	-
3	7 - 8-1/2	6 - 7-1/2	4-1/2 - 6	3 - 4-1/2	3 - 3-1/2
4	9 - 10-1/2	8 - 9-1/2	6-1/2 - 8	5 - 6-1/2	4 - 5-1/2
5	11 - 12-1/2	10 - 11-1/2	8-1/2 - 10	7 - 8-1/2	6 - 7-1/2
6	13 - 14	12 - 13-1/2	10-1/2 - 12	9 - 10-1/2	8 - 9-1/2
7	-	14	12-1/2 - 14	11 - 12-1/2	10 - 11-1/2
8	-	-	-	13 - 14	12 - 14

3.3.2 Fabric and box toe materials.

3.3.2.1 Upper fabric.

3.3.2.1.1 Nylon duck. The cloth for the quarters and gussets shall be nylon duck conforming to class 1 of MIL-C-43734, except that the color shall be Black 375 for type I and Tan 495 for type II, and the air permeability requirement shall not apply. The dimensional stability requirement shall not apply for the type II boot. The nylon duck cloth for the type I only may be treated with a non-fray finish to facilitate cutting and handling. The nylon duck cloth for the collar cover for the type II may be flame-combined to a 0.05 ± 0.008 inch thick layer of polyester foam having a density of 1.70 ± 0.05 pounds per cubic foot. As an alternate, the fabric described in 3.3.2.1.2 may be used.

3.3.2.1.2 Knit lining (type II only). The knit fabric for the quarter and gusset linings shall be made from 100 percent, 70 denier, 34 filament tetra-channel polyester fiber (see 6.6.1). The color shall be light tan approximating the color of the upper leather. The knit material shall be combined with a 0.005 ± 0.008 inch thick layer of polyester foam having a density of 1.70 ± 0.05 pounds per cubic foot. The combined material shall provide a minimum moisture vapor permeability of 1000 grams per square meter per 24 hours when tested as specified in table VIII. The polyester knit/foam lining shall be combined to the nylon duck using a flame system. As an alternate, a knit fabric consisting of 48 percent, 70 denier, 34 filament polyester and 52 percent, 90 denier nylon may be used (see 6.6.1.1). The fabric shall be knit so that the polyester comprises the face and the nylon comprises the back. The color shall be light tan. The knit fabric shall be combined with 0.050 ± 0.008 inch thick layer of polyester foam having a density of 1.70 ± 0.05 pounds per cubic foot. The combined material shall provide a minimum moisture vapor permeability of 1000 grams per square meter per 24 hours when tested as specified in table VIII. The polyester knit/foam lining shall be combined to the nylon duck using a flame system.

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3.3.2.2 Backseam tape. The tape shall conform to type I, class 3, 1/2 or 17/32 inch wide of DDD-T-86. As an alternate, the tape shall conform to type III, class 1, 1/2 inch wide of MIL-T-5038 (see 6.6.2). The color shall be Black 375 for the type I boot and Tan 495 for the type II boot.

3.3.2.3 Box toes. The box toe material shall be a laminate of 0.020 ± 0.002 inch virgin ionomer resin (Surlyn) sandwiched between two layers of non-woven polyester saturated with polystyrene. The total thickness shall be 0.056 ± 0.005 inch. The box toes shall be cut to the patterns provided and shall be skived 3/8 ± 1/8 inch wide at breast. Testing shall be as specified in 4.4.1.

3.3.2.4 Vamp lining. The vamp lining shall be cut from cotton twill conforming to class 2 of MIL-C-43956.

3.3.2.5 Binding tape. The tape for the top edge of the gusset shall be nylon conforming to type III, class 1 or 2, 3/4 inch of MIL-T-5038, except that the color shall be Black 375 for the type I boot and Tan 495 for the type II boot. The dyed tape shall show fastness to weathering equal to or better than the standard sample (see 6.4) when tested as specified in 4.4.1. When no standard sample is available, the dyed tape shall show "good" fastness to weathering when tested as specified in 4.4.1.

3.3.2.6 Quarter top facing, backstay, and ankle reinforcement tape. The type I boot quarter top facing and backstay tape shall be cut from 1 inch nylon webbing, and the ankle reinforcement tape shall be cut from 2 inch or two 1 inch pieces of nylon webbing. The type II boot backstay tape shall be cut from 1 inch webbing. The webbing shall conform to class 1 or 2 of MIL-W-17337, except that the webbing shall be nylon 6,6, and shall be dyed Black 375 for the type I boot and Tan 495 for the type II boot. The dyed webbing shall show colorfastness to weathering equal to or better than the standard sample (see 6.4) when tested as specified in 4.4.1. When no standard sample is available, the dyed webbing shall show "good" fastness to weathering when tested as specified in 4.4.1.

3.3.3 Metal.

3.3.3.1 Steel shank. The steel shank shall be constructed from 19 gage, 0.0418 ± 0.0030 inch cold rolled carbon steel with a hardness ranging from 47 to 54 on the Rockwell C scale and shaped to conform to the arch of the MIL-S last in MIL-L-41815. The width of the shank shall be 1 ± 1/8 inch. As an alternate, the width of the shank for boot sizes 3 through 6-1/2 all widths and 7 through 8-1/2 XN, N, and R widths may be 5/8 ± 1/32 inch. The shank shall be made with three ribs, and the overall height of each rib shall be 0.080 to 0.125 inch. Each rib shall taper off to end within 1 ± 1/8 inch from the toe end and within 1 ± 1/8 inch from the heel end. The shank shall have two prongs, each of 3/16 inch length, upward into the filler and two prongs, each of 3/32 to 5/32 inch length, downward into the insole. The steel shank shall have a zinc electroplated coating with dull or bright finish, or a zinc coating completely and uniformly applied to the base metal

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following a thorough cleansing by any method of TT-C-490. Steel shanks with black oxide coating conforming to class 1 of MIL-C-13924 or steel shanks with zinc phosphate coating may be used in lieu of shanks with zinc coating. The shank length grading shall be as specified in table III. Testing shall be as specified in 4.4.1.

TABLE III. Shank lengths

Boots sizes (all widths)	3 to 6-1/2	7 to 8-1/2	9 to 10-1/2	11 to 12-1/2	13 to 14
Shank length in inches ± 1/32 inch tolerance	4-1/8	4-3/8	4-5/8	4-7/8	5-1/8

3.3.3.2 Tacks and staple wire. The heel seat lasting tacks shall be brass or steel. The brass tacks shall be cut from alloy No. 6 or 8 of ASTM B 36. The side lasting staples and toe lasting staples shall be steel. Tacks shall be of sufficient length to attach the parts through which they are driven and clinch onto the insole.

3.3.3.3 Eyelets. The eyelets shall be gloss finish nylon coated brass conforming to commercial size AA. The eyelets shall be black for the type I boot and Tan 20318 of FED-STD-595 for the type II boot.

3.3.3.4 Drainage eyelets and washers (type I only).

3.3.3.4.1 Drainage eyelets. The drainage eyelets shall be brass conforming to chemical composition copper alloy No. 260 of QQ-B-613. The eyelets shall have a black oxide finish. The drainage eyelet shall have a head with a diameter of 0.345 ± 0.005 inch, the outside diameter of the barrel shall be 0.218 ± 0.002 inch, and the barrel length shall be 0.185 ± 0.005 inch when tested as specified in 4.4.1.

3.3.3.4.1.1 Drainage eyelets (alternate). The drainage eyelets shall be brass conforming to chemical composition copper alloy No. 260 of QQ-B-613. The eyelets shall have a back oxide finish. The drainage eyelets shall have an outside diameter of 0.455 ± 0.005 inch, the outside diameter of the barrel shall be 0.218 ± 0.002 inch, and the barrel length shall be 0.165 ± 0.005 inch.

3.3.3.4.2 Washers. The washers shall be brass conforming to chemical composition copper alloy No. 260 of QQ-B-613. The washers shall have a black oxide finish. The washers shall have an outside diameter of 0.463 ± 0.005 inch, and inside diameter of 0.220 inch, +0.005, -0 inch, and a thickness of 0.009 ± 0.003 inch when tested as specified in 4.4.1.

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3.3.3.5 Steel insole plate (type I only). The steel for the insole plate shall be corrosion-resistant steel strip, composition 301, condition hard, thickness 0.011 ± 0.001 inch, 2D or 2B finish, conforming to MIL-S-5059.

3.3.3.6 Thermal barrier (type II only). The thermal barrier for the type II boot shall be 1/8 inch, plus 1/32 inch polyurethane foam having a density of 22 ± 7 pounds per cubic feet, a tensile strength of 115 pounds per square inch minimum, and a maximum compression set of 22 percent when tested as specified in table VIII (see 6.6.3).

3.3.3.7 Speed lace assembly. The speed lace assembly shall be of two-piece construction consisting of gloss finish nylon coated brass speed lacer and nylon coated alternate full tubular rivet in accordance with Drawing 2-1-1635.

3.3.3.8 Outersole and heel, rubber. The bottom assembly for both DMS biscuit and injection molding methods (rubber outersole and heel) for the types I and II boots shall have the following general characteristics and shall be tested as specified in 4.4.1 and 4.4.4:

Type I:

a. Color: The color shall be black, nonmarking, producing only a slight erasable mark on white paper.

b. Volume swell: The volume swell shall be not greater than 75 percent in medium No. 6 fuel.

c. Abrasive index: The abrasive index shall be not less than 175 before and after accelerated aging for 70 hours at $100^{\circ} \pm 1^{\circ}C$.

d. Hardness: The durometer hardness shall be 60 ± 5. After accelerated aging for 70 hours at $100^{\circ} \pm 1^{\circ}C$, the durometer hardness shall change no more than 10 points from the original.

e. Cut growth: The cut growth, after accelerated aging for 70 hours at $100^{\circ} \pm 1^{\circ}C$, shall be no greater than 100 percent after 50,000 cycles.

f. Ozone resistance: There shall be no sign of cracking after the test for ozone resistance.

g. Bond strength: The bond strength shall be not less than 135 pounds except that one specimen out of the five tested may be below 135 pounds but shall be not less than 120 pounds. If any specimens in the sample show clear evidence of tearing into the leather or outersole rubber below 135 pounds, those specimens shall be discarded and new ones shall be selected for the test.

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h. Specific gravity: The specific gravity shall not be greater than 1.13.

Type II:

a. Color. The color shall be tan, nonmarking, producing only a slight erasable mark on white paper.

b. Volume swell. The volume swell shall be not greater than 75 percent in medium No. 6 fuel.

c. Abrasive index. The abrasive index shall be not less than 175 before and after accelerated aging for 70 hours at 100° ± 1°C.

d. Hardness. The durometer hardness shall be 60 ± 5. After accelerated aging for 70 hours at 100° ± 1°C, the durometer hardness shall change no more than 10 points from the original.

e. Cut growth. The cut growth, after accelerated aging for 70 hours at 100° ± 1°C, shall be no greater than 100 percent after 50,000 cycles.

f. Ozone resistance. There shall be no sign of cracking after the test for ozone resistance.

g. Bond strength. The bond strength shall be not less than 135 pounds except that one specimen out of the five tested may be below 135 pounds but shall be not less than 120 pounds. If any specimens in the sample show clear evidence of tearing into the leather or outersole rubber below 135 pounds, those specimens shall be discarded and new ones shall be selected for the test.

h. Specific gravity. The specific gravity shall not be greater than 1.13.

3.3.3.8.1 Bottom design. The bottom design shall be bars and cleats conforming to Drawing 2-1-962. The sole thickness shall be as specified on the drawing and shall match within 1/16 inch between left and right boots.

3.3.4 Thread.

3.3.5 Thread, upper fitting. The thread for all upper fitting operations shall be bright nylon thread conforming to type I, II, or III, class A or B of V-T-295, except that requirements for colorfastness shall not apply. The nylon thread shall be E for the needle and bobbin with the 301 and 308 stitch type, E for the needle, and B or E for the looper with the 401 stitch type. The color of the thread shall be black for the type I boot and tan C.A. 66019 for the type II boot.

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3.3.6 Thread, insole stitching. The thread for stitching the two layers of the leather insoles shall be a natural color polyester thread conforming to type I, class 1, subclass C, size 5, 3 ply of V-T-285.

3.3.7 Miscellaneous materials.

3.3.7.1 Heel fillers. The heel fillers shall be made of fiberboard or other suitable material that does not affect vulcanization or proper molding of the heel. The fillers shall be heel-shaped with straight edges. The heel fillers shall conform to the dimensions in table IV when tested as specified in 4.4.1.

TABLE IV. Dimensions of heel fillers in inches

Heel filler size designation	A	B	C	D
Heel curve radius	9/16	11/16	13/16	15/16
Width	1-1/8	1-3/8	1-5/8	1-7/8
Length	1-3/4	2	2-1/4	2-1/2
Thickness	19/32	19/32	19/32	19/32
Tolerance	± 1/32	± 1/32	± 1/32	± 1/32

3.3.7.2 Adhesives. The adhesives used for bonding the various leather and fabric components of the boots shall be of the following types:

a. Natural rubber latex
b. Synthetic rubber latex (including chloroprene)
c. Natural rubber solvent cement
d. Synthetic rubber solvent cement
e. Synthetic resin cement

3.3.7.3 Instruction tags. Instruction tags, to be tied to the finished boots, shall be printed using the data supplied in figure 1 for the type I boot, figure 2 for the type II boot, and figure 3 for both boots. In addition, the instruction tags for the type I boot only shall be stamped in red with the words "THIS BOOT IS SPIKE PROTECTIVE" across the top front. The tags shall be printed on tag stock that is 5-1/4 by 6-1/2 inches and folded in the middle to form four printed pages in book form that are 5-1/4 by 3-1/4 inches. A punch hole shall be made in the top near the folded edge for insertion of the lace.

3.3.7.4 Laces. The laces shall conform to type III, class 1 of V-L-61. The color of the laces shall be black for the type I boot and Tan 495 for the type II boot. The requirements for colorfastness to water shall not apply to the type II lace. The length of each lace shall be a minimum of 66 inches for all sizes.

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3.3.7.5 Collar foam (type II only). The foam for the collar shall be 1/4 ± 1/32 inch thick latex foam having a density of 6.7 ± 2 pounds per cubic foot. One side only shall be covered with a cotton scrim cloth (see 6.6.3).

3.3.7.6 Insert insole.

3.3.7.6.1 Type I - ventilating insole. Each pair of boots shall be supplied with one pair of ventilating insoles conforming to type II of MIL-I-1821. The ventilating insole size relation to applicable size boot shall be as indicated in MIL-I-1821.

3.3.7.6.2 Type I - cushion insole (alternate). As an alternate to the ventilating insole, the cushion insole (see 6.6.5) shall be a medium brown open celled polyether or polyester polyurethane foam, with an integral skin, laminated to a neoprene impregnated cellulose substrate (see 6.6.6). The combined thickness of the polyurethane foam cellulose substrate shall be 0.130 ± 0.013 inches. The polyurethane layer shall have a thickness of 0.105 ± 0.013 inches. The cushion insole shall be one-half size less than the boot in the same width. For size 3 in all widths and size 5XN, the insole size shall be the same as the boot. The cushion insole shall conform to the leather insole pattern. A tolerance of +0, -1/16 inch from Government loaned insole patterns will be allowed around the periphery of the insole. The size and width of each insole shall be legibly stamped on the bottom, cellulose substrate, side of the insole in the shank area. The size shall be stamped in numbers and the width in letters. The polyurethane foam shall conform to the following properties:

Properties	Requirements
Density	17 +2, -1 pounds per cubic foot
Tensile strength	115 psi (minimum)
Compression set	10 percent (maximum)
Compression set after hydrolysis 1/	5 percent (maximum)

1/ Less than 5 percent after 5 hours exposure to saturated steam at 15 psi.

3.3.7.6.3 Type II - molded cushion insole. The molded cushion insole (see 6.6.5) shall be a medium brown microcellular urethane foam (see 6.6.6), with an integral skin laminated to a thermo-formed cross-link polyethylene heel cup/arch support. The polyurethane cover shall have a thickness of 0.110 ± 10 percent, and shall include an active organic anti-microbial agent. The top cover shall be perforated. The polyurethane foam shall conform to the following properties:

Properties	Requirements
Density	17 +2, -1 pound per cubic foot
Tensile strength	115 psi (minimum)
Compression set	10 percent (maximum)
Compression set after hydrolysis	10 percent (maximum)

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The size and width of each insole shall be legibly stamped on the bottom (foam side) of the insole in the shank area in numbers and letters not less than 5/16 inch in height. The size of the cushion insole shall be that needed to cover the last bottom of the whole boot and shall be used in such size boot and the next larger half size (size 9R insole to be used in 9R and 9-1/2R). There shall be three widths only, in Regular, Narrow, and Wide. The Narrow shall be used in XN and Wide in XW. The insole shall be cut in such a manner that the full contoured portion of the heel area and arch support will be included. The total thickness of the insole shall be 0.130 to 0.155 inch forward of the instep arch and 0.190 to 0.215 inch elsewhere.

3.3.7.7 Finishing materials for the leather uppers.

3.3.7.7.1 Repairers. Repairers shall be liquid spray, or crayon or paste type applied by hand. The color shall match the upper leather and shall have sufficient coverage to correct minor surface imperfections of the leather.

3.3.7.7.2 Fillers. Fillers, where used, shall be capable of providing a suitable foundation for the application of the top finish and shall be applied by sponge or spray method.

3.3.7.7.3 Top finish. Top finish shall be applied by either the sponge or spray method. It shall be a material compatible with and capable of wetting into and binding the previously applied coats. The finish shall match the dyed leather and shall dry to a low luster.

3.4 Design.

3.4.1 Type I. The design for the type I boot shall be a blucher pattern having a partial speed lace closure and an integral steel sole plate and a removable ventilating or cushion insole. The leather areas on the outside of the boot shall have the grain surface exposed. The boot shall have a nylon cloth quarter with nylon webbing for the backstay, top stay, and ankle reinforcements. The vamp shall be plain with a surlyn box toe, and the outersole shall have bars and cleats in accordance with Drawing 2-1-962 (see 2.1.2). Two drainage eyelets are located on the inner arch area of each boot to facilitate drainage when the boot is submerged in water.

3.4.2 Type II. The design for the type II boot shall be a blucher pattern having a partial speed lace closure and an integral thermal barrier sole piece and removable cushion insole. The leather areas on the outside of the boot shall have the flesh surface exposed. The boot shall have a nylon cloth quarter with leather ankle reinforcement and nylon webbing backstay. The boot shall have a latex foam padded collar covered with nylon cloth. The vamp shall be plain with a surlyn box toe. The boot shall have a lightweight rubber outer sole which shall have bars and cleats in accordance with Drawing 2-1-962 (see 2.1.2). There shall be no drainage eyelets on the vamp.

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3.5 Patterns and dies. The height of the size 9R finished boot, measured upward on outside from tread area at breast of heel to top of boot, shall be 9-7/8 ± 1/4 inches for the type I boot and 10-1/2 ± 1/4 inches for the type II boot and shall graduate up or down between sizes and widths as indicated by patterns. A standard set of paper patterns and markers will be furnished by the Government as a basis from which the contractor's dies and patterns shall be made. The patterns, markers, and contractor's cutting dies and patterns shall consist of the component parts specified in table V. The Government patterns and markers shall not be altered in any way and shall be followed, except for the steel insole plate (see 3.7.8.1.1) and for lasting allowance, which shall be determined by the contractor to provide a minimum of 1/2 inch allowance. As an alternate, the backstay marker may be omitted. An outside counterpocket may be used with an inverted "V" removed in the center of the counterpocket. The height of the cut for this inverted "V" may be as much as 1-1/4 inches from the bottom of the lasting allowance on a size 3 and grading upward to a maximum height of 1-1/2 inches on a size 14. The point of this inverted "V" shall be in the center back of the counterpocket. All outer lines of this counterpocket must conform to Government furnished patterns except for this inverted "V". This "V" shall be removed in a curved line to conform, after closing, to the back part of the MIL-5 last. As an alternate, the type II boots only may be cut up one half size greater than the last size, when needed, for ease of lasting. When this option is utilized, the permanent identification marking (see 3.7.4) shall correspond to the size of the last used. A 1/4 inch seam allowance shall be added to the back of the quarter pattern for the type I boot.

TABLE V. Contractor's dies and patterns

Component parts	Sizes
Vamp	Whole and half sizes, all widths
Quarter (type I)	Whole and half sizes, all widths
Quarter (type II)	Whole and half sizes, all widths
Counterpocket inside	Whole and half sizes, all widths
Counterpocket outside	Whole and half sizes, all widths
Front edge stay (type I)	Whole and half sizes, all widths
Front edge stay (type II)	Whole and half sizes, all widths
Eyelet stay (type I)	Whole and half sizes, all widths
Eyelet stay (type II)	Whole and half sizes, all widths
Gusset, bellows (type I)	Whole and half sizes, all widths
Gusset, bellows (type II)	Whole and half sizes, all widths
Collar cover (type II)	Whole sizes, all widths
Collar foam (type II)	Whole sizes, all widths
Box toe	8 sizes, all widths
Vamp lining	Whole and half sizes, all widths
Insole	Whole and half sizes, all widths
Steel insole plate (type I)	Whole sizes, all widths
Ankle reinforcement (type II)	Whole sizes, all widths

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TABLE V. Contractor's dies and patterns (cont'd)

Component parts	Sizes
Markers	
Vamp	Whole and half sizes, all widths
Quarter (type I)	Whole and half sizes, all widths
Quarter (type II)	Whole and half sizes, all widths
Gusset, bellows (type I)	Whole and half sizes, all widths
Gusset, bellows (type II)	Whole and half sizes, all widths
Ankle reinforcement (type II)	Whole sizes, all widths

3.6 Lasts and molds.

3.6.1 Lasts. The boots shall be lasted on the MIL-5 lasts in MIL-L-41815. All necessary sizes will be loaned to the contractor by the Government.

3.6.2 Molds. The molds shall be made so as to produce an outersole conforming to Drawing 2-1-962 when tested as specified in 4.4.2.1.

3.7 Construction.

3.7.1 Cutting uppers.

3.7.1.1 Cutting leather uppers. The leather components of the uppers shall be cut from grain out leather for the type I boot. The leather for the type II boot components, with the exception of the eyelet stays and inside counterpockets, shall be cut so as to be used flesh out. The vamp for the type I boot and the vamp, front edge stay and ankle reinforcement for the type II boot shall not be split or shaved. No stretchy vamps or parts cut from stretchy leather shall be used. Vamps shall be cut from the bend area of the side of leather.

3.7.1.2 Cutting fabric upper parts. The quarters shall be cut in the warp or filling direction or on an approximately 15 degree bias from the warp direction. The gusset may be cut in the warp, filling, or bias direction. The collar cover for the type II boot shall be cut in the bias direction.

3.7.2 Skiving. The skiving of upper leather parts shall conform to requirements specified in table VI. The eyelet stays, front edge stays, and inside counterpocket shall be split or shaved to the thickness specified in table II. Splitting of the outside counterpocket is optional.

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TABLE VI. Skiving requirements

Part	Location	Scarf width in inches	Scarf edge thickness in millimeters (± 0.3 mm)
Vamp	Throat and wings	1/4 ± 1/16	1.0
Vamp	Toe 1/ —	3/4 ± 1/8	1.5
Outside counter-pocket	Stitch edge and top	3/16 ± 1/32	1.3
Outside counter-pocket	Lasting edge 1/	5/8 ± 1/8	1.5

1/ Skiving of the lasting edge of the toe vamp and outside counterpocket is optional.

3.7.2.1 Finishing leather parts. All exposed leather raw edges shall be stained black for the type I boot using the materials specified in 3.3.7.7.

3.7.3 Crimping. The vamps may be crimped, if necessary, to ensure tight lasting.

3.7.4 Marking, permanent identification. The outersole shall be marked as shown on Drawing 2-1-962. The gusset of each boot shall be marked on the inside, approximately 1 inch from the top center of the gusset, with the correct size and width, the contractor's identification symbol in block form, and the month and year (expressed numerically) of the date of contract. The words "SPIKE PROTECTIVE" shall also be marked on the type I boot only. Marking of case number is mandatory. As an alternate for the type I boot only, the contractor's identification symbol, in block form, and the month and year (expressed numerically) of the date of contract may be embossed on the grain side of the insole in the heel area. The marking shall conform to type IV, class 9 of DDD-L-20. Fastness of the class 9 marking shall be as specified for the class 5 marking. Figures shall be Arabic and the letters shall be Gothic. The figures and letters shall be a minimum of 9/32 inch and a maximum of 3/8 inch in height. The stamping shall be expressed as shown in the following example; alternate formats may be used provided all the data elements are included and requirements are met:

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10 W AB 6----92
SPIKE 1084
PROTECTIVE

3.7.5 Upper fitting. Line marking patterns shall be used for all upper fitting. The upper stitching shall be done with not less than 8 stitches nor more than 10 stitches to the inch. A plus tolerance of 3 stitches per inch will be allowed when stitching over heavy places or turning sharp corners, providing the leather is not damaged. Needles shall be of the proper size for the weight of leather and size of thread used. The back seam shall be joined using FED-STD-751 stitch type 401 for the type I boot and stitch type 308 for the type II boot. All upper fitting except closing of the backseam and assembly of the collar shall be done on a machine using stitch type 301.

3.7.5.1 Fitting ankle reinforcement, quarters, and counterpockets.

3.7.5.1.1 Ankle reinforcement (type I only). When the 2 inch ankle reinforcement is used, it shall be positioned as indicated by the markers and shall be stitched to the exterior of the quarters with three single rows of stitching. Two of the single rows of stitching shall be positioned, one at each edge of the webbing, 1/16 to 3/32 inch from the edge. The third single row shall be positioned in the center of the webbing ± 1/16 inch. When the 1 inch ankle reinforcement is used, the two 1 inch pieces shall be positioned as indicated by markers and shall be stitched to the exterior of the quarters with four single rows of stitching. Four of the single rows of stitching shall be positioned, one at each edge of the 1 inch webbing, 1/16 to 3/32 inch from the edge. The ankle reinforcement shall then be trimmed even with the edges of the quarter.

3.7.5.1.2 Ankle reinforcement (type II only). The leather ankle reinforcement shall be cut in whole sizes only, and shall be used for the designated whole size, and next larger half size. The ankle reinforcement shall be positioned in line with the mark on the quarter and shall be stitched to the exterior of the quarter with three single rows of stitching. Two of the single rows of stitching shall be positioned, one at each edge of the leather, 1/16 to 3/32 inch from the edge. The third single row shall be positioned in the center following the mark provided for this purpose. As an alternate, the top and bottom ends of the ankle reinforcement (between the stitch rows) may be partially or completely stitched, provided that these extra stitches are covered by the overlaid parts. If computer stitching is utilized in such a way as to consistently position the parts and stitch locations, the location of the front edge stay and the outside counter pocket may be indicated by two additional short rows of stitches perpendicular to the center stitch, and the marking of the ankle reinforcement may be omitted.

3.7.5.1.3 Quarters, backstays, and counterpockets (type I only). The nylon duck quarters shall be closed at the back with a closed seam 3/16 to 1/4 inch in width from top to bottom. This seam shall be rubbed down and reinforced with a stitched tape so that the tape will be covered by the backstay. The inside counterpocket shall be stitched grain side out to the

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marked position on the quarters with two rows on the edge. The top edge of the quarter shall be cemented and evenly folded inwards; the fold shall be 1/4 ± 3/32 inch in width. The outside counterpocket shall be stitched to the quarter marker positioned with two double rows of stitching. The individual rows of each double row shall be 1/16 to 1/8 inch apart. The first double row of stitching shall be stitched through the cloth quarter 1/16 to 1/8 inch from the top edge of the outside counterpocket. The second double row of stitching shall be stitched through the outside counterpocket, the fabric quarter, and the inside counterpocket 3/8 ± 1/8 inch from the top edge of the outside counterpocket. The backstay, as specified in 3.3.2.6 shall be positioned 1/4 ± 3/32 inch from the top edge of the quarter, and shall be flush with the edge to plus 1/4 inch beyond the bottom edge of the quarter. The backstay webbing shall be stitched with a double row of stitching along both edges. The outer row of stitching shall be 1/16 to 3/32 inch from the edge. The individual rows of each double row shall be spaced 1/16 to 1/8 inch apart.

3.7.5.1.4 Quarters, backstays, and counterpockets (type II). The nylon duck quarters shall be closed at the back from top to bottom using seam type FSa-1 with a 3/8 ± 1/16 inch wide zigzag stitch type 308, with 8-10 stitches per inch (defined as points per inch). This seam shall be reinforced with stitched tape on the inside of the quarters covering the zigzag stitching. The inside counterpocket shall be stitched grain side out to the marked position on the quarters with two rows on the edge. The top edge of the quarter shall be cemented and evenly folded; the fold shall be 1/4 ± 3/32 inch in width. The outside counterpocket shall be stitched to the quarter marker positioned with two double rows of stitching. The individual rows of each double row shall be 1/16 to 1/8 inch apart. The first double row of stitching shall be stitched through the cloth quarter 1/16 to 1/8 inch from the top edge of the outside counterpocket. The second double row of stitching shall be stitched through the outside counterpocket, the fabric quarter, and the inside counterpocket 3/8 ± 1/8 inch from the top edge of the outside counterpocket. The backstay, as specified in 3.3.2.6 shall be flush with the top edge of the quarter, and shall be flush with the edge to plus 1/4 inch beyond the bottom edge of the quarter. The backstay webbing shall be stitched with a double row of stitching along both edges. The outer row of stitching shall be 1/16 to 3/32 inch from the edge. The individual rows of each double row shall be spaced 1/16 to 1/8 inch apart. The top edge of the backstay webbing shall be singed or otherwise heat-treated to avoid fraying. A ball/round point needle shall be used on all fabric stitches. The needle used shall be the smallest size to prevent needle cutting.

3.7.5.1.4.1 Backstay seam strength (type I only). The seam strength of the backstay shall be not less than 250 pounds per inch except that one specimen out of the five tested may be below 250 pounds per inch but not less than 225 pounds per inch when tested as specified in 4.4.4.

3.7.5.1.5 Quarter top facing (type I only). The quarter top facing, as specified in 3.3.2.6 and cut in one piece, shall be placed even with the top

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edge of the quarter and stitched on the lower and upper edge. The quarter top facing shall overlay the backstay and extend under the outside quarter stay.

3.7.5.1.6 Padded collar (type II only). The padded collar shall be assembled by spot cementing the latex foam (scrim cloth side away from the cover) in the approximate center of the cover. The foam and cover shall be folded horizontally in the center so that the latex foam is snugly enclosed by the cover and secured with a cemented bond along the bottom edge. The width of the cemented bond shall be 5/16 ± 1/16 inch at the bottom edge. As an alternate, the edges of the folded collar may be assembled by means of a stitch type 304, 308, or 404 of FED-STD-751, with 7-9 stitches per inch, positioned to produce a border of the same width as provided by the cemented bond. The assembled collar shall be placed so that the edge of the foam will fit snugly against the edge of the quarter with a double row of stitches 1/16 to 1/8 inch apart.

3.7.5.1.7 Front edge stay (types I and II). The front edge stay shall be placed on the outside of the quarter, grain side out on type I and flesh side out on type II, extending a minimum of 5/16 inch beyond the quarter edge and under the counterpocket, then stitched with one row on the inside edge.

3.7.5.1.8 Inverted "V" seam counterpocket (types I and II). If inverted "V" seam counterpocket is used, the "V" seam shall be closed, sewing upward from the lasting edge. The closing shall be done with one row of stitching using stitch type 401 and stitched with 8-10 stitches per inch. The seam shall extend approximately 1/16 inch on the inside of the counterpocket and shall be rubbed down. The butt side of this seam shall be on the inside of the counterpocket.

3.7.5.2 Gusset fitting, eyelet stay attaching. The top of the nylon and cotton duck gusset shall be bound with the nylon tape specified in 3.3.2.5 and stitched with two rows of stitching. The first row shall be 3/32 ± 1/32 inch from the edge of the tape, and the second row shall be 3/32 ± 1/32 inch from the edge of the gusset; both of these rows of stitching shall be through the gusset. The bottom edge of the gusset shall be cemented and folded toward the inside, the folded part to measure 1/4 ± 3/32 inch in width. The inside leather eyelet stays shall be stitched, grain side out, at the marker position on the gusset, with one row of stitching. The gusset shall be stitched to the vamp with one row of stitching not more than 1/8 inch from the edge of the gusset, and a second row placed not more than 5/32 inch from the edge of the vamp, both of these rows to include stitching through the vamp lining. The gusset shall be stitched to the quarters with two rows of stitching, one row on the outside edge that shall extend not less than 3/8 inch on the quarter top facing (type I), and the other row on the eyelet stay that shall pass through the outer edge of the quarter panel fabric and gusset fabric without superimposing onto the eyelet stay edge row. For the type II, one of the rows shall be secured by means of a three-stitch backtack at the top of the front edge stay, specifically placed to lock any interruption in stitch pattern.

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3.7.6 Vamping and barring. Vamping shall be done with two rows of stitching (single or double needle) close on the edge of the front edge stay, plus two rows spaced approximately 3/16 inch between inside rows. There shall be two single or double bar rows 5/8 ± 1/16 inch in length at each blucher ear. The bar row shall be placed outside and above vamping by 1/16 to 1/8 inch and parallel to the vamping row at blucher point. This barring may be done by the automatic method with tying or locking ends.

3.7.6.1 Inserting drainage eyelets (type I boot). Two drainage eyelets, as specified in 3.3.3.4.1 shall, along with two washers as specified in 3.3.3.4.2, for each drainage eyelet, be assembled and inserted through the vamp on the inside shank of each boot just above the insole on the finished boot. One washer shall be on the grain side of the vamp and the second washer shall be against the vamp lining serving as a base to clinch the eyelet. The hole in the vamp through which the drainage eyelet is inserted shall be no more than 7/32 inch in diameter. As an alternate, two of the drainage eyelets, as specified in 3.3.3.4.1.1 may be inserted from the grain side and each clinched with one drainage washer, as specified in 3.3.3.4.2.

3.7.6.2 Attaching speed lace assembly and eyeletting. On each quarter, there shall be two nylon coated double A eyelets, as specified in 3.3.3.3, for all sizes spaced evenly from blucher nose to the first speed lace loop. The edge of each eyelet shall be 3/16 to 5/16 inch from the edge of the quarter. The type I boot shall have four speed lace loops equally spaced on each quarter above the eyelets for sizes 3 through 6-1/2, five speed lace loops for sizes 7 through 11-1/2, and six speed lace loops for sizes 7 through 14. The type II boot shall have five speed lace loops for sizes 3 through 11 and six speed lace loops for sizes 11-1/2 through 14, spaced in accordance with the marker on each quarter above the eyelets. The speed lace loops shall be placed perpendicular to the edge of the quarters and parallel to each other and shall be securely riveted 1/4 ± 1/32 inch from the center of the rivet to the edge of the quarter.

3.7.7 Lacing for lasting. The machine lacing or hand inserted clip(s) for lasting shall provide for an opening, after lasting, of 3/4 ± 1/8 inch. The quarter shall be laced or clipped in the two lower pairs of eyelets.

3.7.8 Insoles. The insoles shall be sorted and cased for even weight. Insoles shall be fleshed and buffed on the flesh side. Die-cut insoles will be accepted, provided they conform to paper insole patterns loaned by the Government. A tolerance of +0 and -1/16 inch from Government loaned insole patterns will be allowed around the periphery of the insole.

3.7.8.1 Insole construction assembly (type I only). The insoles shall be assembled in three layers. The top layer, with grain surface next to the foot, shall be the 4-iron leather insole. The middle layer shall be the steel insole plate. The bottom layer shall be the 3-iron leather insole.

3.7.8.1.1 Steel insole plate and resistance to penetration. The steel insole plate, as specified in 3.3.3.5, shall be cut such that the distance

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between the outer edge of the plate and the outer edge of the leather insole shall be no more than 3/8 inch when assembled in accordance with 3.7.8.1.2. The Government patterns may be used for guidance (see 3.5). Both surfaces of the steel plate shall be nominally clean. The steel insole plate shall have a minimum average resistance to penetration of 215 pounds when tested as specified in 4.4.1.1. No single determination shall fall below 200 pounds.

3.7.8.1.2 Attaching leather insoles to steel insole plate. The leather insoles, as specified in 3.3.1.2, shall be cemented on the flesh surface of the 4-iron insole and on the top buffed or split surface of the 3-iron insole with any adhesive specified in 3.3.7.2. The leather insoles shall be attached to the steel insole plate, maintaining a margin of approximately 1/4 inch from the edge of the leather insoles to allow for stitching of the two attached layers of the leather insole, and making certain that the 4-iron leather insole grain surface will be against the foot (see 4.4.2.3.1). The assembly shall then be stitched using a polyester thread, with stitch type 101 or 301 of FED-STD-751. The stitching shall be 4 to 6 stitches per inch, shall be not less than 1/16 inch from the edge of the insole, and shall not penetrate the steel (see 4.4.2.3.2). When stitch type 101 is used, the chain shall be placed on the bottom surface of the assembly and not against the foot.

3.7.8.1.3 Insole tack holes and molding. If the insole tack holes are needed (see 3.7.9), prior to lasting, no more than four holes shall be drilled through the assembled insole. The holes shall be no more than 3/64 inch in diameter. The combined leather insole with steel plate may be molded to conform to the contour of the last bottom.

3.7.8.2 Insole construction assembly (type II only). The insoles shall be assembled in three layers. The 4-iron leather insole shall have the grain surface next to the foot. The thermal barrier shall be placed in the middle, and the 3-iron leather insole shall be placed on the bottom.

3.7.8.2.1 Attaching leather insoles to thermal barrier. The leather insoles, as specified in 3.3.1.2, shall be cemented on the flesh surface of the 4-iron insole and on the top buffed or split surface of the 3-iron insole with any of the adhesives specified in 3.3.7.2. The leather insoles shall be attached to the thermal barrier specified in 3.3.3.6 with the edges aligned making certain that the 4-iron leather insole grain surface will face the foot. The assembly shall then be stitched with polyester thread, and using stitch type 101 or 301 of FED-STD-751. The stitching shall be 4 to 6 inches per inch, and shall be not less than 1/16 inch from the edge of the insole. When stitch type 101 is used, the chain shall be placed on the bottom surface of the assembly, as opposed to the surface facing the foot.

3.7.9 Lasting. Insoles of the correct size and width shall be tacked to the last with four tacks. One of the four tacks shall be in the center of the heel seat, one at the shank, one in the ball, and one at the toe

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(see 3.7.8.1.3). As an alternative, the insole may be attached with four tacks on the outside 1/4 inch edge of the insole, so that two of the tacks are at the forepart and two are at the shank. Staples in lieu of tacks may be used for attaching the insole to the last. Care shall be taken that the edge of the insole is flush with or not more than 1/16 inch from the bottom edge of the last at all points and does not extend beyond the last at any point. The counters shall be assembled in accordance with the counter size schedule (see 3.3.1.4) so that they fill the counterpockets. When required for fit, one size smaller may be used. The counter shall be seated to the insole and anchored with one or two tacks near the back end of the counter flange. The tack shall be driven through the outside counterpocket and insole and clinched against the steel insole plate. Steel tacks may be used for lasting operations. The counter shall be well cemented on both sides. When leather counters are used, the flesh side of the leather counter shall be toward the inside of the boot. The counter shall be turned over the last at the bottom 1/2 ± 1/16 inch. Uppers of the correct size and width shall be assembled to the last in a manner to provide for a wiped-in heel seat of not less than 7/16 inch nor more than 3/4 inch. The correct size of the box toe specified in 3.3.2.3 shall be conditioned and inserted between the vamp and vamp lining. A light coating of any adhesive specified in 3.3.7.2 shall be applied between the vamp and vamp lining rearward of the box toe. The uppers shall be drawn over the lasts with the proper tension on the pulling-over machine to ensure that the quarters at the blucher points and the vamps are down to the last and that the blucher noses are even. The heel seat shall be wiped-in and tacked, and shall be flat and free from wrinkles. The toe shall be firmly and smoothly wiped-in and securely attached by staples. The sides shall be cemented before or in conjunction with side lasting and stapled with a side stapling machine. The staples shall be adjusted and spaced to securely hold the upper in place through the vulcanization operation.

3.7.10 Time allowance on lasts. The boots shall remain on the lasts a sufficient length of time to thoroughly dry the counters and uppers. As an alternate, the use of a heat setting machine may be used to thoroughly dry the counters and uppers. The heat setting time shall be approved by the contracting officer.

3.7.11 Tack and staple pulling. All insole tacks and staples shall be removed with care taken that no broken tack or staple points remain.

3.7.12 Preparation of bottom. The bottom shall be prepared in accordance with the procedures detailed in 3.7.12.1 and 3.7.12.2.

3.7.12.1 Roughing. The edges of the lasted uppers shall be roughed with a cylindrical steel brush, 6.25 ± 0.25 inches in diameter and 1 or 2 inches in width, that is revolving at a speed of 2500 ± 400 revolutions per minute. The bristles shall have a diameter of 0.015 ± 0.003 inch. As an alternative, the bottom roughing may be performed with an automatic roughing machine or other suitable automatic roughing equipment. The first stage shall rough the

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outer edge (featherline) to the extent that the roughed surface will mold to the nip line of the outersole mold but will not be evident above the nip line. The second stage shall rough the remainder of the overlay in such a way as to remove the grain and leave the fibers rough and loose. Wheel roughing prior to wire brush roughing may be performed. Care shall be taken not to rough through the overlay. All loose particles of dust shall be completely removed from the roughed bottom with the use of a stiff bristle brush or compressed air.

3.7.12.2 Skiving and roughing. As an alternative to the procedure detailed in 3.7.12.1, the lasted grain side overlay portion of the counterpocket shall be skived in a sufficient width to ensure that the skived surface will mold to the nip line of the outersole mold but will not be evident above the nip line. The scarf edge thickness shall be 1.5 ± 0.3 mm. The entire skived edges, except the area where the stitches overlay the insole, shall be roughed in accordance with 3.7.12.1. The overlay portion of the vamp area shall be roughed in accordance with 3.7.12.1.

NOTE: When this alternative method is used, the skiving of the flesh side of the counterpocket shall not be permitted.

3.7.13 Preparation of upper for molding.

3.7.13.1 For biscuit method. A liberal amount of adhesive shall be applied with a hand brush to the roughed area and to the exposed part of the insole and the shank. After allowing the adhesive to dry for a minimum of 15 minutes, a second coat of the same adhesive shall be applied to the same areas and allowed to dry for a minimum of 40 minutes.

3.7.13.2 For injection method. The adhesive shall be applied in such a manner as to ensure compliance with the bond strength requirements specified in 3.3.3.8.

3.7.14 Attaching steel shank. The proper size of steel shank shall be attached to the upper while the boot is either on the MIL-5 last or on the metal last. The forward end of the steel shank shall be approximately 1-1/8 inches rearward of the ball line.

3.7.15 Relasting. The uppers shall be relasted using the proper size and width of the metal lasts on the direct molded sole machine.

3.7.16 Use of mold release. Coating with a mold release compound will be permitted on those surfaces of the metal mold that will come into contact with the soling compound.

3.7.17 Molding outersole and heel to upper. The outersole and heel shall be direct molded to the upper by either the biscuit method or the injection method (see 6.5), using molding equipment that is approved for this purpose

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by the contracting officer. The metal lasts used in the molding operation shall conform to the measurements contained in MIL-L-41815. The rubber soling compound shall be as specified in 3.3.3.8. The heel fillers shall be as specified in 3.3.7.1 in four sizes (A, B, C, and D) and shall be applied to the boots in accordance with the schedule in table VII..

TABLE VII. Heel filler size schedule

Boot size	Boot width XN	N	R	W	XW
3, 3-1/2	-	-	-	-	A
4, 4-1/2	-	A	A	A	A
5, 5-1/2, 6, 6-1/2	A	A	A	A	A
7, 7-1/2, 8, 8-1/2, 9, 9-1/2	B	B	B	B	B
10, 10-1/2, 11, 11-1/2, 12, 12-1/2	C	C	C	C	C
13, 13-1/2, 14	D	D	D	D	D

3.7.17.1 Biscuit molding method. The biscuits for the outersole and heel sections shall be cut from the same batch of rubber. The outersole biscuit shall extend from the toe to the rear of the heel in one unbroken piece. The heel biscuit shall be made from one or more strips of suitable thickness. A cavity for the heel filler shall be formed or cut in the heel biscuit to allow for a minimum margin of 5/8 inch between the cavity and the outer heel perimeter contacting the lasted upper, not including the heel breast line. The heel biscuit and inner surface of the outersole biscuit may, when necessary, be lightly brushed with methyl ethyl ketone.

3.7.17.1.1 Alternate biscuit molding method. The outersole rubber shall consist of one or more pieces cut or extruded so as to knit together during the molding cycle into a finished sole of the specified dimensions. The heel filler shall be securely keyed to the steel shank, allowing a minimum margin of 5/8 inch between the periphery of the heel filler and the outer perimeter of the lasted upper.

3.7.17.1.2 Biscuit molding conditions. The biscuit molding equipment shall be operated so as to exert a minimum pressure of 250 pounds per square inch (psi) on the compound. The sole mold and side mold shall be heated to a temperature not to exceed 375°F, except that the side mold shall not exceed 300°F in the box toe area. The dwell time at these temperatures during vulcanizing shall be between 12 and 15 minutes. The boots shall be molded with the use of metal lasts which shall conform to the measurements specified in MIL-L-41815.

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3.7.17.2 Injection molding method. The heel filler shall be securely keyed to the steel shank, allowing a minimum margin of 5/8 inch between the periphery of the heel filler and the outer perimeter of the lasted upper. The upper shall be mechanically held against the outersole and the heel mold, and the soling compound shall be injected into the mold under appropriate heat and pressure. The upper shall be held against the mold until the outersole is properly formed and cured.

3.7.18 Heel and edge scouring. After removal from the direct molded sole machine, the flash edges shall be removed when necessary by scouring. The use of a knife for trimming spewed flash is prohibited at the upper leather edge.

3.7.19 Tacks or staples. Tacks, nails, and staples that have been left protruding through or around the insole and cannot be pulled out shall be cut close to the surface and shall leave no protruding stumps.

3.7.20 Upper finishing.

3.7.20.1 Preparation (type I only). The boots shall be cleaned by removing accumulated dirt, such as wax or cement, and the surface of the leather shall be conditioned to receive further application of the finish. All thread ends shall be trimmed.

3.7.20.2 Treeing. All wrinkles shall be removed. No material shall be used that may injure the leather, fabric, or thread.

3.7.20.3 Final finish. The boots shall be repaired and filled. The materials and methods shall be as specified in 3.3.7.7.1 through 3.3.7.7.3.

3.7.21 Lacing. The boots shall be mated. A lace shall be inserted through the top eyelet of the outside quarter of each boot, and the two laces shall be tied firmly together.

3.7.22 Instruction tags.

3.7.22.1 Type I. Instruction tags to be tied to the finished boots shall be printed using the data specified on figures 1 and 3.

3.7.22.2 Type II. Instruction tags to be tied to the finished boots shall be printed using the data specified on figures 2 and 3.

3.7.23 Ventilating insole cushion. One ventilating insole or molded cushion insole, as described in 3.3.7.6, shall be inserted in each boot.

3.7.24 Repairs of stitching for types 301 and 401 upper fitting stitches. Repairs of types 301 and 401 stitches shall be as follows:

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a. When thread breaks or bobbin run-outs occur during stitching, the stitching shall be repaired by restarting the stitching one stitch in back of the end of the secure stitching, except that on the quarter top facing, backstay, ankle reinforcement, and binding tape the stitching shall be restarted not less than 1/2 inch in back of the end of the secure stitching.

b. Thread breaks or two or more consecutive skipped or runoff stitches noted during inspection of the item (in-process or end item) shall be repaired by overstitching. The stitches shall start one stitch in back of the defective area and continue one stitch beyond the defective area onto the existing stitching, except that on the quarter top facing, backstay, ankle reinforcement, and binding tape the stitching shall be not less than 1/2 inch in each direction beyond the defective stitching area.

c. Vamp stitching damaged or missing at the junction of the outersole shall be repaired with a minimum of two stitches. Applicable to each row of defective stitching, the first stitch shall be a maximum of 1/4 inch from the junction of the outersole.

NOTE: In lieu of repairing each row of stitching individually, the four rows of stitching in the vamp may be repaired in one operation by stitching across the rows of stitching. Location of stitching shall be as specified above and shall start and end within the rows of stitching. Each group of four rows of stitching shall be repaired separately.

3.7.25 Replacement of defective components. During the spreading, cutting, and manufacturing process, components having material defects or damages that are classified as defects in 4.4.2.2.1, 4.4.2.2.2, and 4.4.3 shall be removed from production and replaced with nondefective and properly matched components.

3.8 Figures. Figures 1, 2, 3, and 4 are furnished solely for guidance and information. Should variation from the written specification appear in the figures, the written specification shall govern.

3.9 Workmanship. The finished boots shall be free of defects, such as missing components, incorrect stitching patterns, unspecified materials, unmatched pairs, crooked box toe, unspecified color, wrong type or placement of hardware components, protruding points in insole, lack of ventilating or molded cushion insole, or missing instruction label. The boots shall also be free from stains or foreign matter affecting appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by

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the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated nor actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.3 and tested for the characteristics specified in 4.4.4. The presence of any defect or the failure of any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document. In addition, testing shall be performed on components and materials listed in table VIII. When the data items in the "Number of determinations per sample unit" and the "Results reported as" columns are not specified in table VIII, they shall be reported as required by the referenced test methods. A rubber compound lot shall consist of the amount of rubber compound that is produced

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or compounded in one day. A representative composite test sample of the lot of rubber used for the DMS biscuit method shall be vulcanized by using a 10 minute cure at 310°F. A representative composite test sample of the lot of rubber used in the injection molding method shall be molded and cured at the same temperature and time as that used in the production of the boots. The composite test sample shall be as follows:

a. Two pieces, 6 inches by 6 inches by 0.070 ± 0.010 inch.

b. Two pieces, 6 inches by 6 inches by 0.250 ± 0.010 inch.

All test reports shall contain the individual values utilized in expressing the final result. The lot shall be unacceptable if one or more sample units or the composite sample fails to meet any specified requirement except for abrasive index. For abrasive index, all composite samples from the rubber compound lots making up the boot lot shall be tested. If the average of all composite samples fails the abrasive index, the lot of rubber compound shall be rejected. The sample size shall be as follows:

Lot size	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE VIII. Component tests

Component and lot size expressed in terms of	Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Composite	No. determinations per sample unit	Results reported as: Pass or fail	Results reported as: Numerically to nearest	Sample unit
Knit lining (type II only)	Material identification	3.3.2.1.2	1/	-	-	-	-	-	-
	Foam thickness	3.3.2.1.2	D 3574 1/4/	-	-	-	-	-	-
	Foam density	3.3.2.1.2	D 3574 1/4/	-	-	-	-	-	-
	Moisture vapor permeability	3.3.2.1.2	E 96 1/ 4/	-	-	-	-	-	-
Box toes (1 box toe)	Material identification	3.3.2.3	1/	-	-	-	-	-	-
	Impregnation compound identification	3.3.2.3	1/	-	-	-	-	-	-
	Thickness	3.3.2.3	1/	-	-	-	-	-	-
Nylon binding tape (gross yd)	Colorfastness to weathering	3.3.2.5	5671 2/3/	X	-	-	-	-	2 yards
Nylon webbing reinf. tape (gross yd)	Colorfastness to weathering	3.3.2.6	5671 2/3/	X	-	-	-	-	1 yard

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TABLE VIII. Component tests (cont'd)

Component and lot size expressed in terms of	Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Composite	No. determinations per sample unit	Results reported as: Pass or fail	Results reported as: Numerically to nearest	Sample unit
Shank, steel (1 pair)	Material identification	3.3.3.1	1/	-	-	-	-	-	-
	Thickness	3.3.3.1	1/	-	-	-	-	-	-
	Hardness (Rockwell C scale)	3.3.3.1	E 18 4/	X	-	1	-	Hardness number	-
	Height, steel rib	3.3.3.1	Visual, micrometer	X	-	1	X	-	2 shanks
	Length and width	3.3.3.1	1/	-	-	-	-	-	-
	Coating identification	3.3.3.1	1/	-	-	-	-	-	-
	Finish	3.3.3.1	1/	-	-	-	-	-	-
Tacks and staples	Material identification	3.3.3.2	1/	-	-	-	-	-	-
Eyelets 1 pound	Material identification	3.3.3.3	1/	-	-	-	-	-	-
	Finish	3.3.3.3	1/	-	-	-	-	-	-
Drainage eyelets 1 pound (type I only)	Diameter head	3.3.3.4.1	Gage	X	-	2	X	-	-
	Outside diameter of barrel	3.3.3.4.1	Gage	X	-	2	X	-	-

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TABLE VIII. Component tests (cont'd)

Component and lot size expressed in terms of	Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Composite	No. determinations per sample unit	Results reported as: Pass or fail	Results reported as: Numerically to nearest	Sample unit
Drainage eyelets 1 pound (type I only) (cont'd)	Length of barrel	3.3.3.4.1	Gage	X	-	2	X	-	2 drainage eyelets
	Outside diameter	3.3.3.4.1.1	Gage	X	-	2	X	-	-
	Length overall	3.3.3.4.1.1	Gage	X	-	2	X	-	-
Washers (type I only)	Outside diameter	3.3.3.4.2	Gage	X	-	2	X	-	2 washers
	Inside diameter	3.3.3.4.2	Gage	X	-	2	X	-	-
	Thickness	3.3.3.4.2	Gage	X	-	2	X	-	-
Thermal barrier (type II only)	Thickness	3.3.3.6	1/	-	-	-	-	-	-
	Density	3.3.3.6	D 3574 1/4/	-	-	-	-	-	-
	Tensile strength	3.3.3.6	D 3574 1/4/	-	-	-	-	-	-
	Compression set	3.3.3.6	D 3574 1/4/	-	-	-	-	-	-
Speed lace loops	Material identification	3.3.3.7	1/	-	-	-	-	-	-
Outersole and heel compound (lot)	Volume swell	3.3.3.8	6001 and 6211 5/	-	X	3	-	One percent	-
	Abrasive index (before and after aging)	3.3.3.8	D 1630 4/ 6/ 7/	-	X	3	-	Whole number	-

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TABLE VIII. Component tests (cont'd)

Component and lot size expressed in terms of	Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Composite	No. determinations per sample unit	Results reported as: Pass or fail	Results reported as: Numerically to nearest	Sample unit
Outersole and heel compound (lot) (cont'd)	Hardness (before and after aging)	3.3.3.8	D 2240 4/ 7/	-	X	5	-	Whole number	-
	Cut growth (after aging)	3.3.3.8	D 1052 4/ 6/	-	X	2	-	One percent	-
	Ozone resistance	3.3.3.8	4.5.1 1/	-	X	1	X	-	-
	Specific gravity	3.3.3.8	14011 8/	-	X	3	-	.001	-
Heel filler (1 filler)	Heel curve radius	3.3.7.1	Gage	X	-	2	X	-	-
	Width	3.3.7.1	Gage	X	-	2	X	-	2 fillers of each size in the lot
	Length	3.3.7.1	Gage	X	-	2	X	-	
	Thickness	3.3.7.1	Gage	X	-	2	X	-	
Collar foam (type II only)	Material identification	3.3.7.5	1/	-	-	-	-	-	-
	Thickness	3.3.7.5	D 1055 1/4/	-	-	-	-	-	-
	Density	3.3.7.5	D 1055 1/4/	-	-	-	-	-	-
Insole, cushion alternate (1 square yard material)	Material identification	3.3.7.6.2	1/	-	-	-	-	-	-
	Thickness	3.3.7.6.2	1/	-	-	-	-	-	-
	Density	3.3.7.6.2	D 3574 1/4/ (Test A)	-	-	-	-	-	-

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TABLE VIII. Component tests (cont'd)

Component and lot size expressed in terms of	Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Composite	No. determinations per sample unit	Results reported as: Pass or fail	Results reported as: Numerically to nearest	Results reported as: Sample unit
Insole, cusion alternate (1 square yard material) (cont'd)	Tensile strength	3.3.7.6.2	D 3574 1/4/ (Test E)	-	-	-	-	-	-
	Compression set	3.3.7.6.2	D 3574 1/4/ (Test D)	-	-	-	-	-	-
	Compression set after hydrolysis	3.3.7.6.2	D 3574 1/4/ (Test D)	-	-	-	-	-	-
Contoured cushion insole	Type/material identification	3.3.7.6.3	1/	-	-	-	-	-	-
	Thickness	3.3.7.6.3	D 3574 1/4/ 9/ or D 3575	-	-	-	-	-	-

1/ A certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ Refers to FED-STD-191.

3/ The time of exposure shall be 40 standard hours.

4/ Refers to ASTM Standard.

5/ Refers to FED-STD-601.

6/ Aging shall be at a temperature of $100^{\circ} \pm 1^{\circ}C$ for 70 hours in accordance with the ASTM D 573 procedure for accelerated aging.

7/ The abrasive paper to be obtained from the Carborundum Co. shall be G40E155F.

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8/ Specific gravity calculation to be obtained as a by product of the volume swell test from the same specimens.

9/ If all polyethylene foam use D 3575; otherwise use D 3574.

4.4.1.1 Testing of steel insole plate (type I only). The steel insole plate shall be tested as specified in 4.5.2 for conformance to the resistance to penetration requirements specified in 3.7.8.1.1. Fifteen insole plates shall be selected at random from the lot and tested. Any test failure shall be cause for rejection of the lot of insole plates.

4.4.2 In-process inspection. Inspection shall be made at any point or during any phase of manufacturing to determine whether operations or assemblies are accomplished as specified. When the DMS biscuit method is used, the contractor shall furnish a certificate of compliance for the dwell time. The Government reserves the right to exclude from consideration for acceptance any material or service for which in-process inspection has indicated nonconformance.

4.4.2.1 Mold approval. During the molding operation and prior to the formation of the initial inspection lot, one boot shall be randomly selected from the production of each new mold and determination shall be made whether the outersole conforms to the measurements specified on Drawing 2-1-962. If the outersole is found to be in conformance, the mold from which it was made shall be approved. If the measurements are found to be nonconforming, an additional five boot outersoles produced from that mold shall be measured for conformance to Drawing 2-1-962 and, if all five outersoles conform to the measurements, the mold shall be approved. If all molds used in the processing of the initial inspection lot have been approved, testing of the finished boots for conformance to outersole measurements shall not be applicable. When new molds are added or when molds are repaired, mold approval must be obtained as described previously.

4.4.2.1.1 Identification marking of approved molds. Each mold, after approval, shall have an identifying number inscribed approximately in the center of the heel breast portion of the heel mold. The mold identification numbering shall be consecutive, beginning with number "1". The height of the numbers shall be 1/8 to 3/16 inch.

4.4.2.2 In-process examination. The defects found during in-process examination shall be classified in accordance with 4.4.2.2.1 and 4.4.2.2.2. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major and 6.5 for total (major and minor combined) defects.

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NOTE: Defects in tables IX, X and XII designated by an asterisk shall be scored as major when seriously affecting serviceability and minor when affecting serviceability but not seriously.

4.4.2.2.1 Examination of uppers after all fitting. The completely fabricated upper assembly prepared for lasting shall be examined for defects that cannot be seen in the end item, as listed in table IX. The lot size shall be expressed in units of one upper assembly. The sample unit shall be one upper assembly.

TABLE IX. Upper defects

Examine	Defect	Classification	
		Major	Minor
Construction and workmanship (general)	Any component missing or not specified type	101	
	Any component misplaced or not affixed as specified	*	
	Vamp throat or wings not skived	102	
	Counterpocket not skived as specified		201
	Vamp stitched more than 5/32 inch from edge at throat	103	
	Vamp stitched more than 3/32 inch from edge at throat but not over 5/32 inch		202
	Backseam not rubbed down		203
Quality of leather	Thickness more than 1/2 ounce less than the minimum specified	104	
	Thickness less than specified minimum but not exceeding 1/2 ounce		204
	Thickness more than the maximum specified		205
	Off-stretch cut	105	
	Slaughter cut	*	

4.4.2.2.2 Examination of boot before last pulling. The partially fabricated boot assembled to the point just prior to pulling off the MIL-5 last shall be examined for defects that cannot be seen in the end item, as listed in table X. The lot size shall be expressed in units of one partially fabricated boot. The sample unit shall be one partialy fabricated boot.

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TABLE X. Boot before last pulling defects

Examine	Defect	Classification	
		Major	Minor
Bottom of boot	Any component missing or not specified type	101	
	Upper damaged		*
	Insole not fleshed and buffed on the flesh side as required	102	
	Poor heel seat, side, or toe lasting		*
Upper part of boot	Upper not tightly pulled down to last		*
	Lace opening less than 5/8 inch or more than 7/8 inch		201
	Quarter not laced as specified		202
	Less than 1/2 inch of upper lasted over insole except heel seat upper less than 7/16 inch or more than 3/4 inch	103	
	Upper in lasted area not sufficiently roughed or cleaned after roughing	104	
	Excessive roughing or scouring		*

4.4.2.3 Examination of the assembled insole (type I only).

4.4.2.3.1 Steel plate. Each assembled insole shall be examined to ensure that the steel plate has not been omitted. A mechanical or manual metal detector or other suitable automatic metal detecting equipment shall be used. As an alternative to metal detecting equipment, the assembled insole may be hand flexed to ensure that the plate has not been omitted. Any assembled insole that does not include the steel insole plate shall be rejected.

4.4.2.3.2 Insole assembly. the assembled insole shall be examined for defects listed in table XI. The lot size shall be expressed in units of one assembled insole. The sample unit shall be one assembled insole. The inspection level shall be II, and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

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TABLE XI. Insole defects

Examine	Defect
Insole stitching	More or less than the specified number of stitches Wrong stitch type Less than 1/16 inch from the edge of the insole Stitches penetrating steel insole Skipped or broken stitch
Steel plate	Distance between outer edge of steel plate and outer edge of leather insole more than 3/8 inch (examined at ball and toe areas)

4.4.3 End item visual examination. The finished boots shall be examined for the defects in table XII. The lot size shall be expressed in units of one boot. The sample unit shall be one completely fabricated boot, and the selection shall be by pairs. The inspector shall check to see that insoles are inserted in the boots. For the pairing examination and when determining a possible difference in outersole thickness, the pair shall be examined together. Each defect found during the examination for pairing shall be scored as a single defect. The inspection level shall be II, and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major and 6.5 for total (major and minor combined) defects. To examine the vamp for break, the boot shall be held in an upright position with both hands. The toe of the boot shall face away from the examiner. Position thumbs on top of vamp approximately halfway between box toe line and blucher noses and 1 inch to 1-1/2 inches apart. For the type I boot only, press downward with thumbs so as to form grain surfaces into a concave surface and observe the break pattern. Any vamp exhibiting a break pattern at any location comparable to a break pattern represented by numbers 5 through 8 on the Satra scale (see 6.7) shall be scored as a defect. The defect shall be scored regardless of the direction in which the break pattern appears.

NOTE: Appearance defects shall be scored only when the condition is plainly visible at a distance of 3 feet or more. Defects designated by an asterisk shall be scored as major when seriously affecting serviceability or appearance and minor when affecting serviceability or appearance but not seriously.

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TABLE XII. End item visual defects

Examine	Defect	Classification	
		Major	Minor
Pairing	Not properly mated (i.e., not right and left of same size)	101	
	Variation in color, luster, or appearance		*
	Variation of more than 1/4 inch in height of pair	102	
	Box toe crooked, long, or short		*
	Difference in outersole thickness between left and right boots:		
	-more than 3/32 inch	103	
	-more than 1/16 inch but not more than 3/32 inch		201
Cleanness	Any nonremovable spot, stain, or foreign matter affecting appearance		202
Color and finish	Not specified color	104	
	Finish streaky, chipped, or flaky on upper		203
Design, type, and size	Not as specified	105	
Upper leather	Leather deeply snuffed (i.e., fiber structure damaged)	106	
	Any grub or tick damage, scratch, brand, or boney leather		*
	Fat wrinkles, flanky leather, or veins		*
	Stretchy vamp	107	
	Leather printed	108	
Break pattern of vamp	Break pattern comparable to break pattern represented by numbers 5 through 8 on Satra break scale	109	
	NOTE: Leather exhibiting a break pattern comparable to the numbers indicated above is considered to be loose, pipey, or flanky. The break pattern shall be determined in accordance with 4.4.3.		

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TABLE XII. End item visual defects (cont'd)

Examine	Defect	Classification	
		Major	Minor
Nylon quarter and gusset	Any mend or weak spot	110	
	Two or more yarns missing	111	
	Any quarter cut on the bias	112	
	Any abrasion, spot, stain, or discoloration		*
Construction and workmanship	Any cut, tear, hole, repair, abrasion, or factory damage		*
	Any component or assembly omitted or misplaced, operation omitted or not properly performed (unless otherwise classified herein)		*
Webbing	Not firmly and tightly woven, edges frayed or scalloped	113	
Seams and stitching (upper)	Open seam not repaired	114	
	Repair of open seam not as specified		204
	Quarter panel fabric and gusset fabric not caught in second row of inside eyelet stay stitching		205

NOTE: A seam shall be classified as open when two or more consecutive skipped or run-off stitches occur; or when one or more stitches joining a seam are broken, except that where vamp stitching crosses the counterpocket stitching area or the quarter top facing, backstay, ankle reinforcement and binding tape, two or more broken stitches shall constitute an open seam. On multiple stitched seams, a seam is considered open when either one side or both sides of seams are open. When any of the above conditions occur on the inside of the boot, it shall be scored as an asterisk (*) defect.

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TABLE XII. End item visual defects (cont'd)

Examine	Defect	Classification	
		Major	Minor
Vamp stitching at outersole junction	One or more rows of defective stitching not repaired	115	
	Repaired but first stitch more than 1/4 inch from outersole junction	116	
	Repaired with one stitch in lieu of two or more stitches		206
	Tight tension resulting in puckering or cutting of leather or fabric	*	
	Loose tension resulting in a loosely secured seam	*	
	Wrong stitch or seam type	117	
	Any row of stitching with less than 8 stitches per inch but not less than 6 stitches per inch		207
	Any row of stitching with less than 6 stitches per inch	118	
	More than the specified maximum number of stitches	*	
	NOTE: A plus tolerance of 3 stitches per inch will be allowed when stitching over heavy places or turning sharp corners.		
	Gage of stitching not as specified or irregular	*	
	Stitching omitted where required	*	
	Thread ends not trimmed throughout boot		208
	Needle holes or needle chew	*	
	Sand hole (i.e., tongue not properly caught in at throat)		209
Outersole and heel	Thickness not as specified	119	
	Flash edges not removed by scouring, affecting appearance		210
	Edge scouring is irregular, affecting appearance		211
	Blister:		
	-1/2 inch or more in longest dimension	120	
	-penetrating 1/3 or more the thickness of the rubber	121	

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TABLE XII. End item visual defects (cont'd)

Examine	Defect	Classification	
		Major	Minor
Outersole and heel (cont'd)	More than three blisters more than 1/8 inch but less than 1/2 inch in longest dimension	122	
	One, two, or three blisters more than 1/8 inch but less than 1/2 inch in longest dimension		212
	Any void or knit mark 3/4 inch or more in longest dimension	123	
	Any void or knit mark penetrating 1/3 or more the thickness of the rubber	124	
	More than one void or knit mark in the outside sole lugs or rear heel lug, 1/2 inch or more in longest dimension	125	
	More than three voids or knit marks in the outersole and heel, 1/2 inch or more in longest dimension	126	
	One, two, or three voids or knit marks 1/2 inch up to but not including 3/4 inch in longest dimension		213
	More than two voids or knit marks more than 1/4 inch but less than 1/2 inch in longest dimension		214
	Any other defect in tread or edge area	*	
	Incorrect or incomplete vulcanization of outersole or heel to uppers (i.e., rubber not securely attached)	*	
Counters	Rolled or curled counter	*	
	Soft counter		215
Heel finishing	Heel not finished square and smooth, affecting appearance		216
Eyelets, quarter	Number of eyelets not as specified but each row has the same number		217
	Not the same numer of eyelets in each row	127	

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TABLE XII. End item visual defects (cont'd)

Examine	Defect	Classification	
		Major	Minor
Eyelets, quarter (cont'd)	Eyelets not properly spaced within the row of misalignment between the rows to an extent interfering with proper lacing		*
	Edge of any eyelet less than 1/16 inch from edge of quarter	128	
	Edge of any eyelet less than 3/16 inch but not less than 1/16 inch or more than 5/16 inch from the edge of the quarter		218
	Any eyelet not securely clinched		219
Drainage eyelets and washers (type I only)	Any drainage eyelet or washer missing	129	
	Not securely clinched, or not inserted as specified		*
Gusset	Binding missing	130	
Barring	Missing	131	
Insole	Short or long		*
	Any protruding point of lasting staple, or tack in heel 1/	132	
Fabric: quarters vamp lining, or ankle reinforcement	Vamp lining or quarter and ankle reinforcement having excessive fullness; torn, loose, or wrinkled		*
Ventilating or molded cushion insole	Missing, wrong size, or wrong side up		220
Markings and instruction tag	Missing, incomplete, incorrect, not applied in the specified manner, misplaced, illegible, or not specified size		221
Speed laces	Number of speed lace loops not as specified but each row has the same number		222
	Not the same number of loops in each row	133	

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TABLE XII. End item visual defects (cont'd)

Examine	Defect	Classification	
		Major	Minor
Speed laces (cont'd)	Loops not properly spaced within the row or misaligned between the rows to an extent interfering with proper lacing		*
	Loops not perpendicular to edge of quarters or parallel to each other		223
	Any loop not securely riveted	134	
	Distance from center of rivet to edge of quarter less than 7/32 or more than 9/32 inch		*
	Four splits in rivet clinch roll		224
	Five or more splits in rivet clinch roll	135	
Color and finish	Any raw edges not stained to match upper leather		225

1/ Any protruding point of lasting staple or tack forward of heel breast line found in the sample shall be cause for rejection of the lot represented.

4.4.4 End item testing. The finished boot shall be tested for the characteristics listed in table XIII. The sample unit for all tests shall be one boot. All test results shall contain the individual values utilized in expressing the final result. The lot shall be unacceptable if one or more sample units fail to meet any requirement specified. The sample size shall be five boots regardless of lot size.

TABLE XIII. End item tests

Characteristics	Requirement paragraph	Test method	No. determinations per sample unit	Results reported as Pass or fail	Numerically to nearest
Nonmarking (outersole and heel)	3.3.3.8	4.5.3	1	X	-
Bond strength (outersole)	3.3.3.8	4.5.4	1	-	lb.

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TABLE XIII. End item tests (cont'd)

Characteristics	Requirement paragraph	Test method	No. determinations per sample unit	Results reported as Pass or fail	Results reported as Numerically to nearest
Measurements 1/	Drawing 2-1-962	Gage	1	X	-
Backstay seam strength	3.7.5.1.4.1	4.5.5	1	-	lb.

1/ Mold approval (see 4.4.2.1).

4.4.5 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted, incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container Unit pack not as specified (when applicable)
Content	Number per container is more or less than required

4.4.6 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

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Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

4.5 Methods of inspection.

4.5.1 Ozone resistance test. There shall be no sign of cracking of the outersole compound when tested as follows: Test specimens 1 inch in width, 3-3/4 inches in length, and 0.070 ± 0.010 inch in thickness shall be mounted in accordance with the requirements of ASTM D 518, procedure B, except that the length of the clamping strips shall be such as to facilitate placement within the test chamber of the ozone cabinet. Samples shall be exposed for 7 days to an ozone concentration of 50 ± 10 parts per hundred million parts of air at a temperature of $40^{\circ} \pm 1^{\circ}C$ as described in ASTM D 1149. At the end of the exposure time, the specimens shall be examined under a seven-power magnifier.

4.5.2 Resistance to penetration test.

4.5.2.1 Apparatus. Resistance to penetration shall be performed on a testing machine having a movable platform adjusted to travel at 1/4 inch per minute. Two blocks of hard wood, metal, or plastic shall be prepared as follows: The blocks shall be of such size and thickness as to ensure a suitably rigid test ensemble and allow for at least 1 inch of the pointed end of an 8D common nail to be exposed for the penetration. One block shall have a hole drilled to hold an 8D common nail firmly at an angle of 90 degrees. The second block shall have a maximum 1/2 inch diameter hole drilled through it so that the hole will allow free passage of the nail after it penetrates the insole plate during the test.

4.5.2.2 Procedure. Place a new 8D nail in the hole of the block, a sample of leather outersole stock 10 to 11 irons thick superimposed above the nail, the area of the insole plate to be tested on the outersole, and the second block with hole so as to allow for free passage of the nail after it passes through the outersole stock and insole plate in that order. Start the machine and record the pressure required for the nail to completely penetrate the outersole and insole plate, to the nearest 5 pounds. Make two determinations on each insole plate and average the results. A new nail shall be used for each determination. Any average reading below 215 pounds or any single determination falling below 200 pounds shall be considered a test failure.

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4.5.3 Test for nonmarking outersoles and heels. The edge of the outersole or heel shall be drawn across a sheet of white bond paper using moderate pressure. If only a light mark is made that is easily erased by rubbing with the fingers, the outersole or heel shall be classed as nonmarking.

4.5.4 Bond strength test of outersole to leather.

4.5.4.1 Specimen. The test specimen shall be a single finished boot of any size prepared as follows: The boot shall be legibly marked on each side at two points (i.e., at 2 inches and 4 inches from the tip of the toe). The molded outersole shall then be separated from the upper to the 2 inch mark. The separation may be aided by any suitable device or machine capable of withstanding the amount of force necessary to accomplish separation. The use of pincers conforming to type II, class 1 of GGG-N-350 or equivalent, with the sharp cutting edge ground down to a 1/8 inch face, and a last one or two sizes smaller than the boot, inserted into the boot and mounted on a last stand, have been found to facilitate the preparation of specimens for the bond strength test. After separation to the 2 inch mark is obtained, the boot last, when used, shall be removed and the toe cap of the boot shall be crushed to facilitate mounting of the specimen in the tester. In the event that separation at the 2 inch mark is not between the upper and rubber outersole, that portion of the rubber outersole remaining adhered to the upper shall be separated manually with the aid of a knife or other instrument so as to show separation between leather and rubber outersole. Care should be exercised so that the leather is not cut.

4.5.4.2 Procedure. The apparatus for measuring outersole-upper separation shall be as described in Method 2031 of FED-STD-311. Grip the separated and crushed toe portion of the boot by one clamp in a manner such that the minimum effective jaw surface area applied shall be 1 inch by 1 inch. (With some testing machines, modification of the clamps may be required to provide sufficient opening for insertion of the specimen.) Center the jaw surface approximately 1/2 to 1 inch from the tip of the toe. The other jaw surface shall hold the overlay portion of the separated upper and shall be not less than 1 inch by 2 inches with the long dimension perpendicular to the pull of the machine. Grip the separated rubber outersole portion of the boot by the other clamp of the testing machine. Each jaw of the clamp shall be not less than 1 inch by 2 inches with the long dimension perpendicular to the pull of the machine. Set the machine in operation, and continue the separation at a speed of 10 $\pm$ 2 inches per minute until the outersole is separated from the upper to a point past the mark 4 inches from the tip. The maximum value attained during separation shall be taken as the bond strength.

4.5.5 Backstay seam strength test.

4.5.5.1 Specimen. The same boot used for the bond strength test may be used for this test. The test specimen shall be a single finished boot of any size prepared as follows: The fabric quarters shall be cut from the leather

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upper and edge stay and cut down the center of the gusset. A straight line shall be drawn on the specimen perpendicular to the seam, 1-1/2 to 2 inches above the bottom of the backseam.

4.5.5.2 Procedure. The calibrated apparatus for measuring backstay seam strength shall be as described in Method 2031 of FED-STD-311, except that the face jaw shall be 1 inch by 1 inch and the back jaw shall be 1 inch by 3 inches. The jaw faces shall be serrated metal to prevent slippage in the clamps. The distance between the clamps shall be 3 inches at the start of the test. The jaw speed shall be 10 to 12 inches per minute. Insert the specimen marked as described in 4.5.5.1 in the clamps so that the seam is centered between the jaws with the line placed in the center of the jaw faces and set the machine in operation. The maximum value attained during separation shall be taken as the backstay seam strength.

5. PACKAGING

5.1 Preservation. Preservation shall be level A.

5.1.1 Level A.

5.1.1.1 Unit pack. Each pair of boots shall be unit packed in a one-piece die-cut fiberboard set-up box. The boots shall be separated from each other by tissue paper conforming to type I of A-A-1249. The box shall be constructed from domestic corrugated single-wall minimum grade 200 fiberboard in accordance with PPP-F-320. Approximate inside dimensions shall be as specified in table XIV.

5.1.1.2 Alternate unit pack. When specified (see 6.2), each pair of properly mated boots shall be tied together by the ends of laces which have been inserted through the top speed lace of the outside quarter of each boot.

5.2 Packing. Packing shall be level A or B, as specified (see 6.2).

5.2.1 Level A packing.

5.2.1.1 Six pack. Six pairs of boots of one size and width only, unit packed as specified in 5.1.1.1, shall be packed in a snug-fitting shipping container conforming to style RSC, grade V2s of PPP-B-636. Unit packs shall be arranged one across the length, two across the width, and three in depth. The approximate inside dimensions of the shipping container shall be as specified in table XIV. Each container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping except for type II, class F loads.

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TABLE XIV. Unit pack and shipping container dimensions

Approximate inside dimensions of unit pack and shipping container

Range of boot sizes	Unit pack dimensions	Shipping container dimensions	Pallet number
3 through 5	10-1/2 x 9-1/2 x 5	19-1/2 x 11 x 15-1/2	102
5-1/2 through 8	11-1/2 x 10 x 5	20-1/2 x 12 x 15-1/2	102
8-1/2 through 11	12-1/2 x 11 x 5-1/2	23 x 13-1/2 x 15-3/8	91
11-1/2 through 14	14 x 11-1/2 x 5-1/2	24 x 14-3/8 x 15-3/8	3

5.2.1.2 Alternate six pack. When specified (see 6.2), six pairs of boots of one size and width only, preserved as specified in 5.1.1.2, shall be arranged as shown in table XV and packed in a fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636, except that the abutting ends of the liner do not need to be joined. The approximate inside dimensions of the shipping container shall be as specified in table XV. A sheet of 30-pound minimum basis weight kraft paper conforming to A-A-203 shall be placed between each layer of boots in order to protect the leather from damage by the speed lace loops. A sheet of this paper shall also completely cover the top and bottom of the contents of each container. Each container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

TABLE XV. Alternate six pack dimensions and packing arrangements

(A) Approximate inside dimensions of shipping container 1/

Range of boot sizes	Length, inches	Width, inches	Depth, inches	Pallet number
3 through 5	23-1/2	12	11	91
5-1/2 through 8	24-1/2	13	12	91
8-1/2 through 11	25-1/2	13-3/4	13	91
11-1/2 through 14	27	14-1/2	14	6

1/ Approximate dimensions are furnished as a guide only. For boot sizes below 5XN and above 12XW, the contractor shall determine the size

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container to use. The dimensions of the container shall be such that it will result in a snug fit of the contents to be packed therein. The use of blocking materials to accommodate small size boots in an oversize container will not be permitted.

(B) Packing arrangements. Arrangement of boots within the shipping container shall be in three layers with two pairs per layer. Each pair shall be in a flat position, laces to laces, with soles in opposing directions. The first two pairs of boots shall be placed end-to-end. The third pair shall be placed on top of the second pair also end-to-end. The fourth pair shall be placed on top of the first pair in the same manner as the third pair was positioned to the second pair. Each layer of boots shall be alternated in opposite direction as the previous layer.

5.2.1.3 Alternate twelve pack. When specified (see 6.2), twelve pairs of boots of one size and width only, preserved as specified in 5.1, shall be arranged as shown in table XVI(B) and packed in a fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636, except that the abutting ends of the liner do not need to be joined. The approximate inside dimensions of the shipping container shall be as specified in table XVI(A). A sheet of 30-pound minimum basis weight kraft paper conforming to A-A-203 shall be placed between each layer of boots in order to protect the leather from damage by the speed lace loops. A sheet of this paper shall also completely cover the top and bottom of the contents of each container. Each container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

TABLE XVI. Twelve pack shipping container dimensions and packing arrangements

(A) Approximate inside dimensions of shipping container 1/

Range of boot sizes	Length, inches	Width, inches	Depth, inches	Pallet number
3 through 7-1/2	30	14-1/2	11-1/2	94
8 through 11-1/2	31	15-1/2	13-1/2	94
12 through 14	34	17	15	1

1/ Approximate dimensions are furnished as a guide only. For boot sizes below 5XN and above 12XW, the contractor shall determine the size container to use. The dimensions of the container shall be such that it will result in a snug fit of the contents to be packed therein. The use

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of blocking materials to accommodate small size boots in an oversize container will not be permitted.

(B) Packing arrangements. Arrangement of boots within the shipping container shall be in three layers with four pairs per layer. Each pair shall be in a flat position, laces to laces, with soles in opposing directions. The first two pairs of boots shall be placed end-to-end in the length of the container. The third pair shall be placed on top of the second pair with the heels next to the toe of the second pair. The fourth pair shall be placed on top of the first pair in the same manner as the third pair was positioned to the second pair. The first four pairs of boots constitute one layer. Each layer of boots shall be alternated in opposite direction as the previous layer (see figure 4).

5.2.2 Level B packing.

5.2.2.1 Six pack. Six pairs of boots of one size and width only, preserved as specified in 5.1.1.1, shall be packed in a fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Unit packs shall be arranged one along the length, two across the width, and three in depth. The approximate inside dimensions of the shipping container shall be as specified in table XIV. Each container shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.2.2.2 Alternate six pack. When specified (see 6.2), six pairs of boots of one size and width only, preserved as specified in 5.1.1.2, shall be arranged as shown in table XV(B) and packed in a fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636, except that the abutting ends of the liner do not need to be joined. The approximate inside dimensions of the shipping container shall be a specified in table XV(A). A sheet of 30-pound minimum basis weight kraft paper conforming to A-A-203 shall be placed between each layer of boots in order to protect the leather from damage by the speed lace loops. A sheet of this paper shall also completely cover the top and bottom of the contents of each container. Each container shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.2.2.3 Alternate twelve pack. When specified (see 6.2), twelve pairs of boots of one size and width only, preserved as specified in 5.1.1.2, shall be arranged as shown in table X(B) and packed in a fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636, except that the abutting ends of the liner do not need to be joined. The approximate inside dimensions of the shipping container shall be as specified in table X(A). A sheet of 30-pound minimum basis weight kraft paper conforming to A-A-203 shall be placed between each layer of boots in order to protect the leather from damage by the speed lace loops. A sheet

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of this paper shall also completely cover the top and bottom of the contents of each container. Each container shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.2.2.4 Weather-resistant shipping container. When specified (see 6.2), the shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of PPP-B-636.

5.3 Palletization. When specified (see 6.2), boots packed as specified in 5.2.2 shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147 except that the use of a sheet of paper or fiberboard between every other layer is not required. Pallet type shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III or IV of MIL-STD-731. Each prepared load shall be bonded with primary or secondary straps in accordance with bonding means C and D or film bonding means F or G. Pallet pattern number shall be as specified in table XIV, XV or XVI, as applicable.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but it is not mandatory).

6.1 Intended use. The type I boots are intended for use in hot, wet tropical areas and the type II boots are intended for use in hot, dry desert areas by military personnel of the Department of Defense.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type, size, and width required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.3, and 6.3).
e. When an alternate unit pack is required (see 5.1.1.2).
f. Selection of applicable levels of packing (see 5.2).
g. Type and class of unit load required (see 5.2.1).
h. When an alternate six pack is required for level A packing (see 5.2.1.2).
i. When an alternate twelve pack is required for level A packing (see 5.2.1.3).

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j. When an alternate six pack is required for level B packing (see 5.2.2.2).
k. When an alternate twelve pack is required for level B packing (see 5.2.2.3).
l. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.4).
m. When palletization is required (see 5.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

6.4 Standard sample. For access to samples, address the contracting activity issuing the invitation for bids or request for proposal, or as directed by the contracting activity.

6.5 Direct molding methods for attaching soles (see 3.7.17).

6.5.1 Biscuit method. Slabs (biscuits) of compounded uncured rubber are placed in heated molds. The biscuits are applied against prepared uppers under heat and pressure necessary to form the outersoles and heels and to vulcanize them to the uppers.

6.5.2 Injection method. Prepared uppers are mechanically held against outersole and heel molds. The soiling compound is then injected into the molds under heat and pressure and held against the uppers until the outersole and heel are formed and cured.

6.6 Sources of supply.

6.6.1 Suggested source for knit lining. A suggested source for polyester lining meeting the requirements of 3.3.2.1.2 is Starensier Inc. (Mesh Knit Fog Coolmax), 5 Perkins Way, P.O. Box 408, Newburyport, MA 01950-0508.

6.6.1.1 Suggested source for alternate knit lining. A suggested source for the alternate knit lining meeting the requirements of 3.3.2.1.2 is Faytex Corporation (mesh knit DRI-LEX Style Number 867), 185 Libbey Parkway, Weymouth, MA 02189.

6.6.2 Suggested source for alternate backseam tape. A suggested source for the alternate backseam tape meeting the requirements of 3.3.2.2 is Bally Ribbon Mills, Bally, PA.

6.6.3 Suggested source for thermal barrier. A suggested source of supply for thermal barriers meeting the requirements of 3.3.3.6 is Foamtex Corporation (Custom Felt 10-700CE), Eddystone, PA.

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6.6.4 Suggested source for collar foam. Suggested sources for collar foam meeting the requirements of 3.3.7.5 are: Foam Products Corporation (Latex E-70, 1/4 inch), 350 Beamer Rd., S.W. Calhoun, GA 30701 or Cushman Marden (Exclusive Agent for Ludlow composites, Freemont, OH, Product Number 2806), 56 Pulaski St., P.O. Box 3001, Peabody, MA 01960.

6.6.5 Suggested source for cushion insoles. Suggested sources for cushion insoles meeting the requirements of 3.3.7.6.2 or 3.3.7.6.3 are: Rogers Foam, 145 Central Street, Somerville, MA 02145, Gemini Laminating Corporation (Product Number 139300), 35 Balch Street, Building K, P.O. Box 398, Beverly, MA 01915 or Frelonic (Product Number FI-9503), 63 Grove Street, P.O. Box 169, Salem, MA 01970.

6.6.6 Suggested source for microcellular urethane foam. A suggested source for the microcellular foam meeting the requirements of 3.3.7.6.2 or 3.3.7.6.3 is Rogers Foam, Poron Division, East Woodstock, CT 06244.

6.7 Satra scale. The Satra scale may be obtained from the British Shoe and Allied Trade Research Association, Satra House, Kettering, Northants, England or may be obtained from Satra House's North American agent: Bata Engineering Batawa, Ontario, Canada.

6.8 International standardization agreements. Certain provisions of this document are the subject of international standardization agreement as cited in NATO, STANAG No. 2333. When amendment, revision, or cancellation of this document is proposed that will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

6.9 Subject term (key word) listing.

Combat
Desert
Dry
Shoe
Tropical
Wet

6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Custodians:

Army - GL
Navy - NU
Air Force - 99

Review activities:

Army - MD
Navy - MC
Air Force - 82
DLA - CT

User activity:

Air Force - 45

Preparing activity:

Army - GL

(Project 8430-0405)

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INSTRUCTION TAG (FOR TYPE I)

BOOTS, HOT WEATHER

INSTRUCTIONS FOR USE

1. Wear with: 1 Pair socks, wool, cushion sole.
1 Pair insoles.

2. Sizes. The boot is supplied in whole and half sizes 3 through 14 in all widths, except sizes 3XN and 3-1/2XN, 3N and 3-1/2N, 3R and 3-1/2R, 3W and 3-1/2W, and 4XN and 4-1/2XN. In some instances it may be necessary to make a size adjustment in order to obtain an adequate fit due to normal swelling of the feet and the use of ventilating insoles.

3. Pull on boot, seating heel firmly into place, then lace. Boots should fit snugly, but not tightly. There should be approximately 1/2 inch additional length at toe.

4. Carry extra socks with you. Change socks often, at least once daily. Remove boots, and dry feet as often as possible. Neglect of the feet can cause Jungle Rot and make you a casualty.

5. Remove insoles, wash insoles in warm (not boiling) soapy water and shake out to dry.

6. Brush mud and excessive dust off boots. To prevent leather from getting hard and stiff, wax often with shoe polish containing silicone.

7. Clean screened vents so that water will drain out.

8. Wipe inside of boot occasionally with a moist, soapy rag, and let dry.

9. As frequently as possible, dry boots and insoles in the sun. Do not place near intense heat, as this will destroy nylon, leather, and plastic.

10. Before putting boots in snake, insect, scorpion, or spider infested areas, carefully inspect the inside of the boots to avoid painful and disabling bites and stings.

FIGURE 1. Hot weather boots instruction tag (type I only)

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INSTRUCTION TAG (FOR TYPE II)

BOOTS, HOT WEATHER

INSTRUCTIONS FOR USE

1. Wear with: 1 Pair socks, wool, cushion sole.
1 Pair insoles.

2. Sizes. The boot is supplied in whole and half sizes 3 through 14 in all widths, except sizes 3XN and 3-1/2XN, 3N and 3-1/2N, 3R and 3-1/2R, 3W and 3-1/2W, and 4XN and 4-1/2XN. In some instances it may be necessary to make a size adjustment in order to obtain an adequate fit due to normal swelling of the feet and the use of insoles.

3. Pull on boot, seating heel firmly into place, then lace. Boots should fit snugly, but not tightly. There should be approximately 1/2 inch additional length at toe.

4. Carry extra socks with you. Change socks often, at least once daily. Remove boots, and dry feet as often as possible. Neglect of the feet can make you a casualty.

5. Brush mud and excessive dust off boots. These boots have a sueded surface and are not designed to be polished.

6. Wipe inside of boots occasionally with a moist, soapy rag, and let dry.

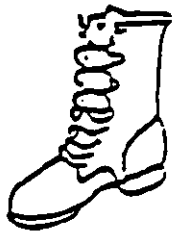
7. As frequently as possible, dry boots and insoles in the sun. Do not place near intense heat as this will destroy nylon and leather.

8. Before putting on boots in snake, insect, scorpion, or spider infested areas, carefully inspect the inside of the boots to avoid painful and disabling bites and stings.

FIGURE 2. Hot weather boots instruction tag (type II only)

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THE FOLLOWING ARE LACING INSTRUCTIONS
FOR SPEED LACES



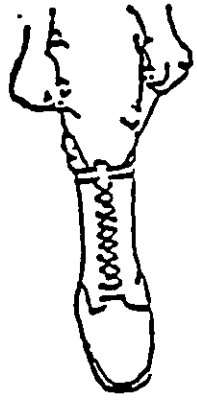
STEP 1

After first lacing through all loops and tying each end of the lace with a hard knot, loosen up lace for easy step-in of foot.



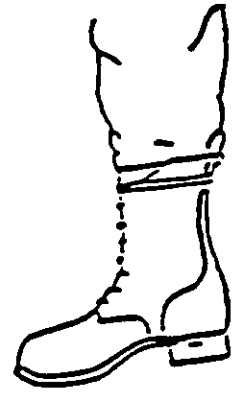
STEP 2

With thumbs on either edge, remove all pleats and wrinkles by pushing tongue securely around leg.



STEP 3

Hold lace ends at pre-tied knots. Pull up lace to desired tightness.



STEP 4

Wrap extra lace length around collar and secure.

FIGURE 3. SPEED LACE INSTRUCTIONS

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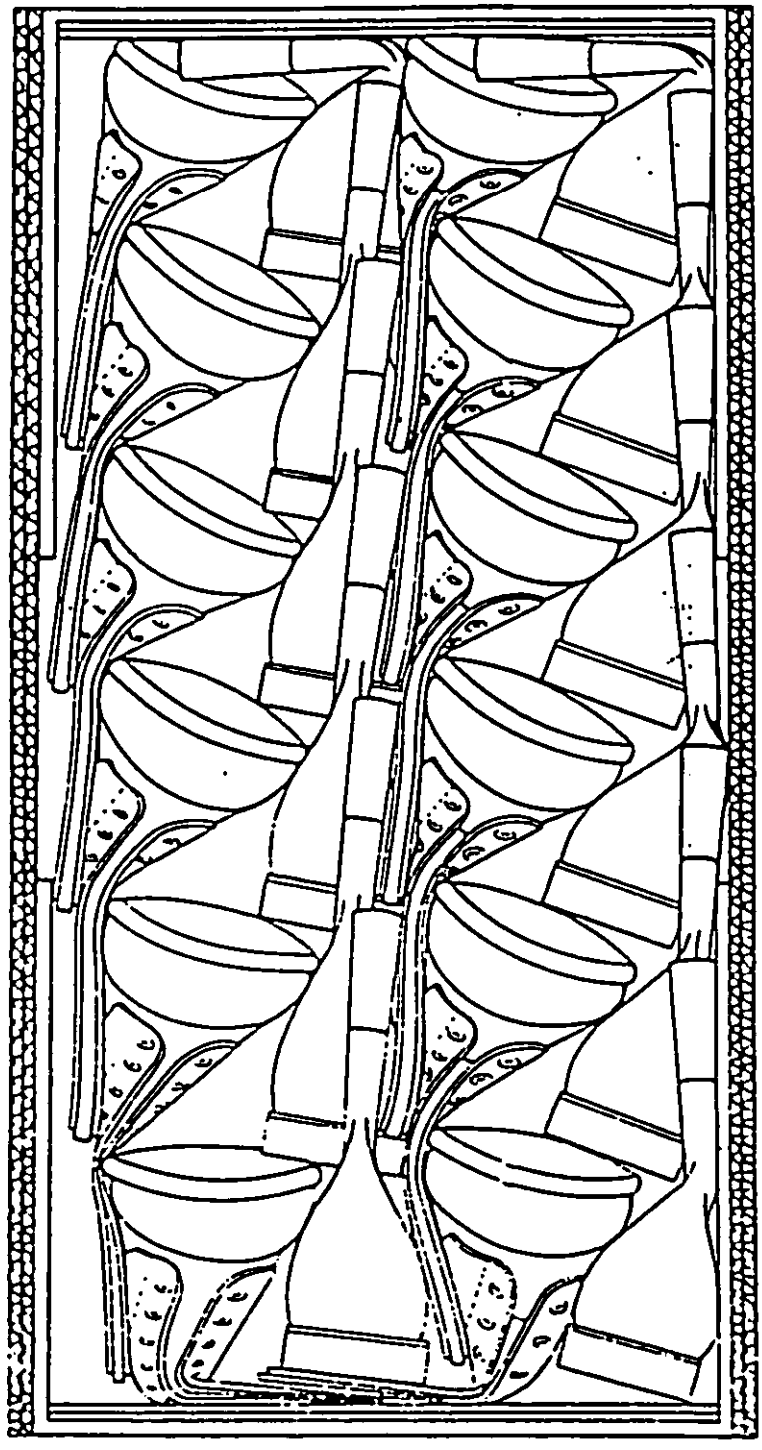


FIGURE 4. BOOT ARRANGEMENT WITH SHIPPING CONTAINER (SIDE REMOVED)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-B-43154M	1992 June 26

3. DOCUMENT TITLE

BOOTS, HOT WEATHER

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
c. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial (2) AUTOVON (*If applicable*)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (*Include Area Code*) (1) Commercial	(2) AUTOVON/DSN
U.S. Army Natick RD&E Center	508-651-4532	256-4532
c. ADDRESS (*Include Zip Code*) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-IRT Natick, MA 01760-5019	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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